

计算机网络实验报告

实验 6 —— OSPF 拓展实验

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	姓名	学号	评分（按百分制）
组长：	王胜伟	23336228	100
组员 1：	宋信贤	23336207	100
组员 2：	王一澄	23336233	100
组员 3：	林宏宇	23320093	100

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1 拓展实验一：单区域 OSPF、认证与故障模拟

1.1 实验要求

本实验要求在给定的四路由器拓扑下，完成以下任务：

- (a) 在所有路由器上配置单区域 OSPF 协议，并增加区域认证功能，最终实现主机 PC1 与 PC2 之间可以相互 ping 通。
- (b) 设计并实施有效方法（如端口镜像），以捕获任意两个路由器之间交换的 OSPF 路由信息报文，并进行分析。
- (c) 模拟 R3 与 R4 之间冗余链路中断的故障，通过命令行工具观察并记录 OSPF 的路由信息变化以及数据传输路径的自动切换过程。

1.2 实验拓扑与 IP 规划

本次实验采用下图所示的四路由器冗余拓扑，笔者使用 17 组与 15 组的设备完成此次实验。详细的物理接线与 IP 规划如下。

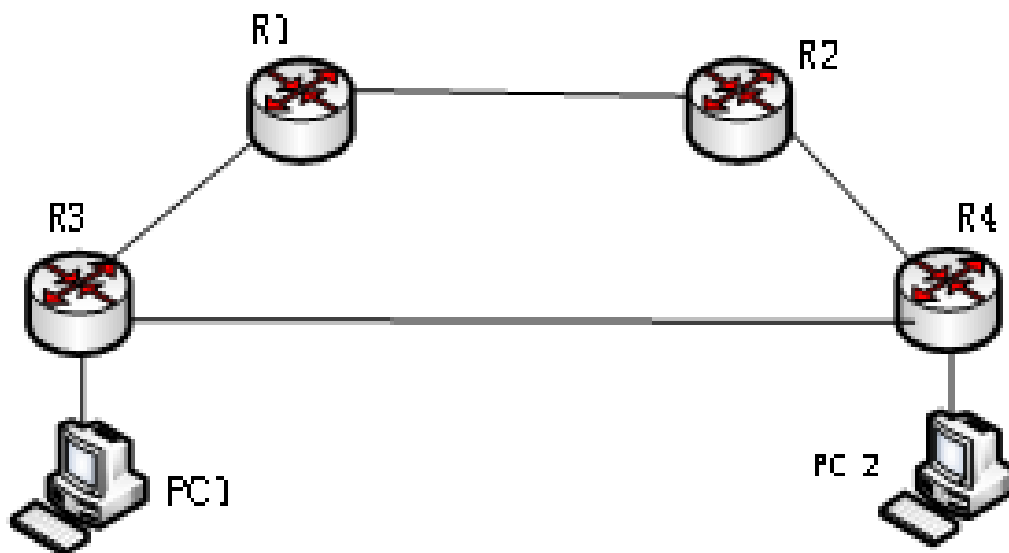


图 1. 拓展实验 1 网络拓扑图

• 物理接线:

- PC1(实验网卡) ↔ R3 G0/0/1
- PC2(实验网卡) ↔ R4 G0/0/7

- R3 G0/0/0 ↔ R1 G0/0/0
- R1 G0/0/1 ↔ R2 G0/0/7
- R2 G0/0/5 ↔ R4 G0/0/5
- R3 G0/0/3 ↔ R4 G0/0/8
- PC3(校园网卡) ↔ R3 G0/0/2 (抓包观察口)

• IP 规划:

- PC1-R3: 192.168.1.0/24
- PC2-R4: 192.168.2.0/24
- R3-R1: 192.168.13.0/24
- R1-R2: 192.168.12.0/24
- R2-R4: 192.168.24.0/24
- R3-R4: 192.168.34.0/24

1.3 实验步骤与结果分析

1.3.1 第一步：基础配置与初始验证

操作描述: 按照拓扑图完成物理接线，为所有 PC 和路由器接口配置 IP 地址，构建网络基础骨架。

```
system-view
sysname R1
interface GigabitEthernet0/0/0
 ip address 192.168.13.2 24
 undo shutdown
 quit
interface GigabitEthernet0/0/1
 ip address 192.168.12.1 24
 undo shutdown
 quit
```

Listing 1. R1 接口 IP 配置命令

```
2025-12-15 20:55.22 /home/mobaxterm telnet 172.16.15.203
Connected to 172.16.15.203.
Entering character mode
Escape character is '^]'.

Login authentication

Username:user
Password:
<AR6140-15-1>system-view
Enter system view, return user view with Ctrl+Z.
[AR6140-15-1]sysname R1
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]ip address 192.168.13.2 24
[R1-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[R1-GigabitEthernet0/0/0]quit
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]ip address 192.168.12.1 24
[R1-GigabitEthernet0/0/1]undo shutdown
Info: Interface GigabitEthernet0/0/1 is not shutdown.
[R1-GigabitEthernet0/0/1]quit
[R1]quit
<R1>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 2. R1 接口 IP 配置过程

```
system-view
sysname R2
interface GigabitEthernet0/0/7
ip address 192.168.12.2 24
undo shutdown
quit
interface GigabitEthernet0/0/5
ip address 192.168.24.1 24
undo shutdown
quit
```

Listing 2. R2 接口 IP 配置命令

```
2025-12-15 20:59.13 /home/mobaxterm telnet 172.16.15.204
Connected to 172.16.15.204.
Entering character mode
Escape character is '^]'.

Login authentication

Username:user
Password:
<AR2200-15-2>system-view
Enter system view, return user view with Ctrl+Z.
[AR2200-15-2]sysname R2
[R2]interface GigabitEthernet0/0/7
[R2-GigabitEthernet0/0/7]ip address 192.168.12.2 24
[R2-GigabitEthernet0/0/7]undo shutdown
Info: Interface GigabitEthernet0/0/7 is not shutdown.
[R2-GigabitEthernet0/0/7]quit
[R2]interface GigabitEthernet0/0/5
[R2-GigabitEthernet0/0/5]ip address 192.168.24.1 24
[R2-GigabitEthernet0/0/5]undo shutdown
Info: Interface GigabitEthernet0/0/5 is not shutdown.
[R2-GigabitEthernet0/0/5]quit
[R2]quit
<R2>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 3. R2 接口 IP 配置过程

```
system-view
sysname R3
interface GigabitEthernet0/0/1
 ip address 192.168.1.1 24
 undo shutdown
 quit
interface GigabitEthernet0/0/0
 ip address 192.168.13.1 24
 undo shutdown
 quit
interface GigabitEthernet0/0/3
 ip address 192.168.34.1 24
 undo shutdown
 quit
```

Listing 3. R3 接口 IP 配置命令

```
2025-12-15 20:55.30 /home/mobaxterm telnet 172.16.17.203
Connected to 172.16.17.203.
Entering character mode
Escape character is '^'.

Login authentication

Username:user
Password:
<AR6140-17-1>system-view
Enter system view, return user view with Ctrl+Z.
[AR6140-17-1]sysname R3
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1]ip address 192.168.1.1 24
[R3-GigabitEthernet0/0/1]undo shutdown
Info: Interface GigabitEthernet0/0/1 is not shutdown.
[R3-GigabitEthernet0/0/1]quit
[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]ip address 192.168.13.1 24
[R3-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[R3-GigabitEthernet0/0/0]quit
[R3]interface GigabitEthernet0/0/3
[R3-GigabitEthernet0/0/3]ip address 192.168.34.1 24
[R3-GigabitEthernet0/0/3]undo shutdown
Info: Interface GigabitEthernet0/0/3 is not shutdown.
[R3-GigabitEthernet0/0/3]quit
[R3]quit
<R3>quit

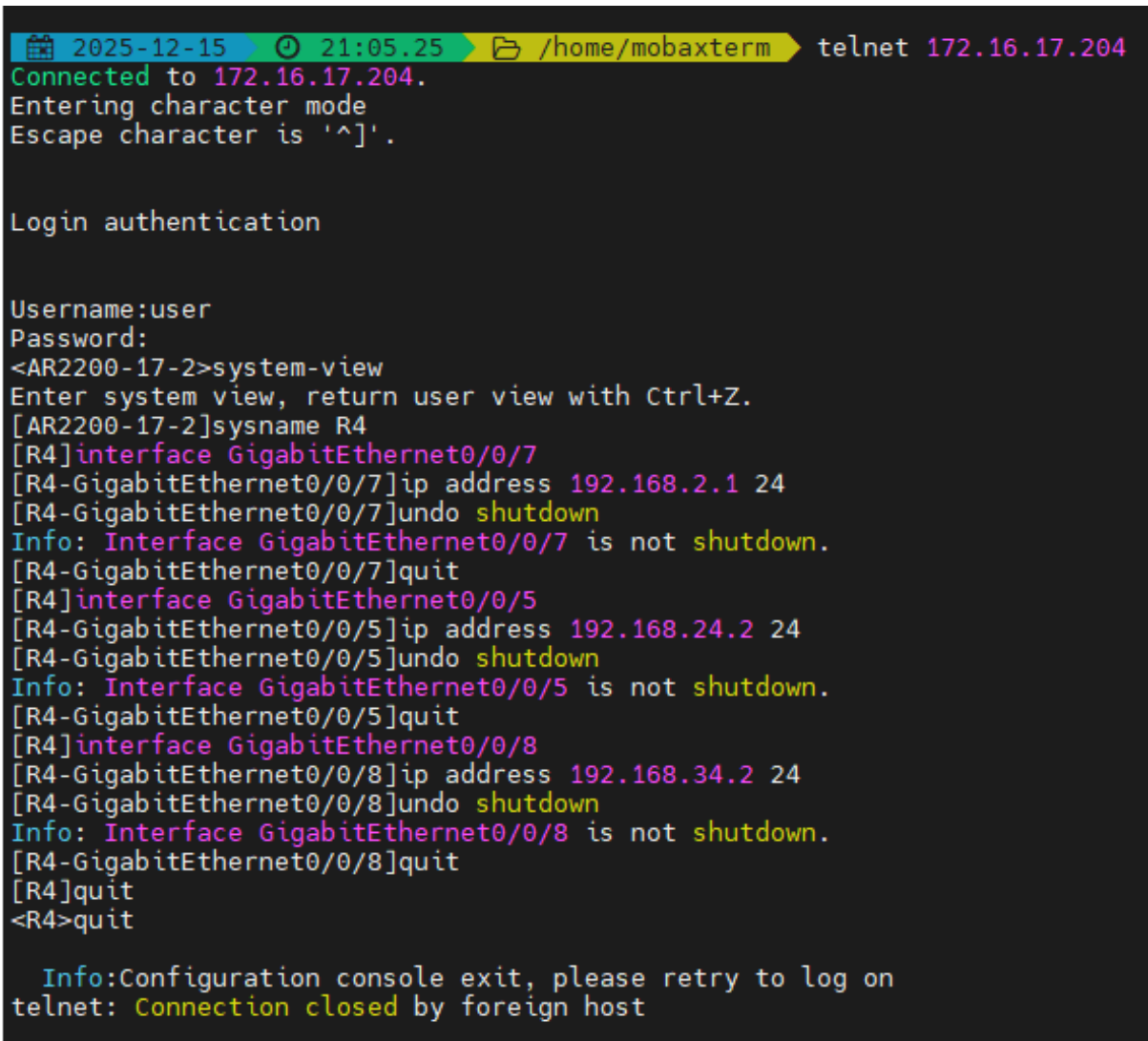
Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 4. R3 接口 IP 配置过程

```
system-view
sysname R4
interface GigabitEthernet0/0/7
ip address 192.168.2.1 24
undo shutdown
quit
interface GigabitEthernet0/0/5
ip address 192.168.24.2 24
undo shutdown
quit
interface GigabitEthernet0/0/8
ip address 192.168.34.2 24
undo shutdown
```

```
quit
```

Listing 4. R4 接口 IP 配置命令



```
2025-12-15 21:05.25 /home/mobaxterm telnet 172.16.17.204
Connected to 172.16.17.204.
Entering character mode
Escape character is '^]'.

Login authentication

Username:user
Password:
<AR2200-17-2>system-view
Enter system view, return user view with Ctrl+Z.
[AR2200-17-2]sysname R4
[R4]interface GigabitEthernet0/0/7
[R4-GigabitEthernet0/0/7]ip address 192.168.2.1 24
[R4-GigabitEthernet0/0/7]undo shutdown
Info: Interface GigabitEthernet0/0/7 is not shutdown.
[R4-GigabitEthernet0/0/7]quit
[R4]interface GigabitEthernet0/0/5
[R4-GigabitEthernet0/0/5]ip address 192.168.24.2 24
[R4-GigabitEthernet0/0/5]undo shutdown
Info: Interface GigabitEthernet0/0/5 is not shutdown.
[R4-GigabitEthernet0/0/5]quit
[R4]interface GigabitEthernet0/0/8
[R4-GigabitEthernet0/0/8]ip address 192.168.34.2 24
[R4-GigabitEthernet0/0/8]undo shutdown
Info: Interface GigabitEthernet0/0/8 is not shutdown.
[R4-GigabitEthernet0/0/8]quit
[R4]quit
<R4>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 5. R4 接口 IP 配置过程

初始连通性测试: 在配置任何路由协议之前，我们对 PC1 和 PC2 分别进行端到端 ping 测试，结果不通，符合预期，因为数据包在路由器上因缺少路由信息而被丢弃。


```
C:\Users\Administrator>ping 192.168.2.10

正在 Ping 192.168.2.10 具有 32 字节的数据:
请求超时。
请求超时。
请求超时。
请求超时。

192.168.2.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 0, 丢失 = 4 (100% 丢失),
```

图 6. 初始连通性测试: PC1 ping PC2 失败

```
C:\Users\Administrator>ping 192.168.1.10

正在 Ping 192.168.1.10 具有 32 字节的数据:
请求超时。
请求超时。
请求超时。
请求超时。

192.168.1.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 0, 丢失 = 4 (100% 丢失),
```

图 7. 初始连通性测试: PC2 ping PC1 失败

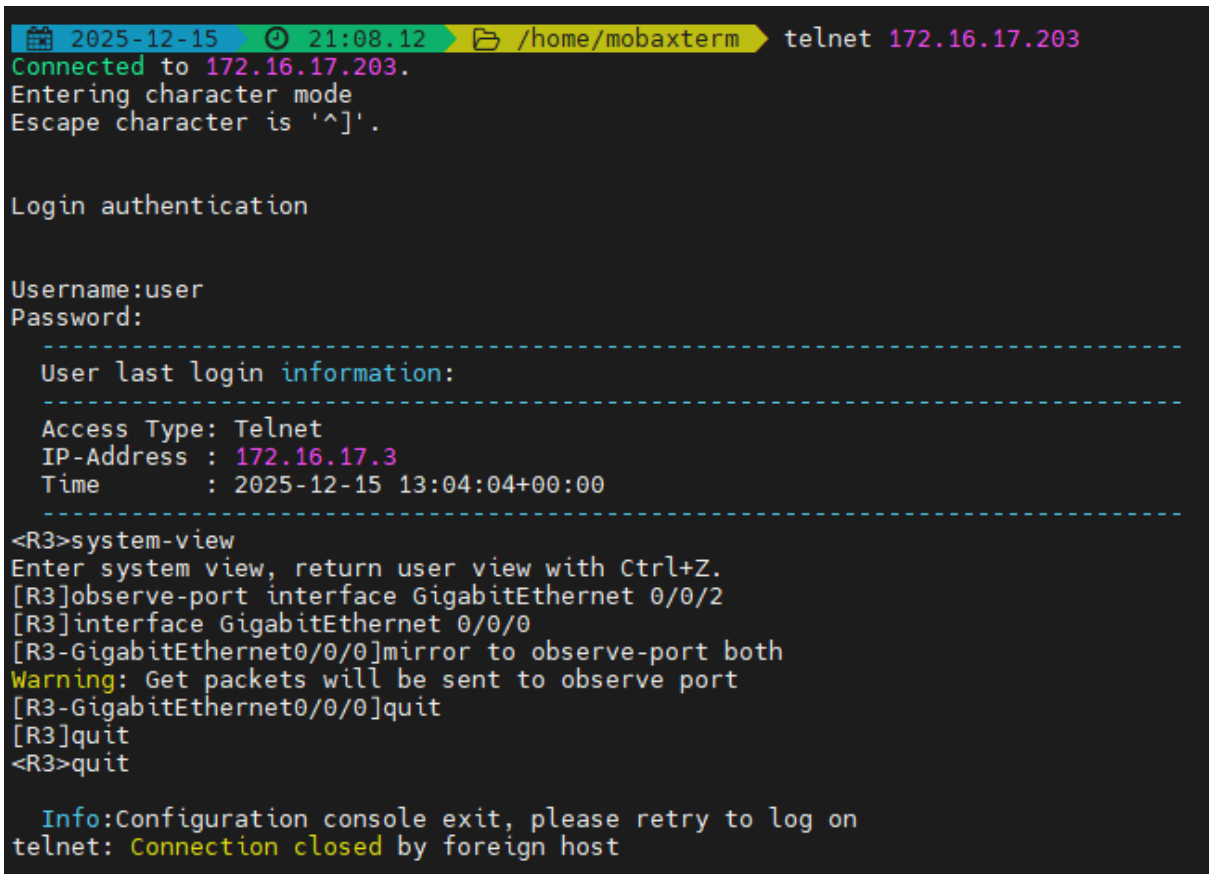
1.3.2 第二步: 配置端口镜像 (任务 b 准备)

操作描述: 为了捕获路由器之间的 OSPF 报文, 我们在 R3 上配置端口镜像。其原理是将一个或多个“镜像端口”(被监听口)的流量复制一份, 发送到指定的“观察端口”。我们将 R3 连接 R1 的 G0/0/0 接口设为镜像端口, 将连接 PC3 的 G0/0/2 接口设为观察端口。(如果想观察其他的路由器之间的路由信息交换报文, 可以采用完全类似的方式来配置镜像并观察。)

```
system-view
! 定义G0/0/2为观察端口
observe-port interface GigabitEthernet 0/0/2
! 进入被监听的接口G0/0/0 (连接R1)
interface GigabitEthernet 0/0/0
! 将双向流量镜像到观察端口
mirror to observe-port both
```

```
quit
```

Listing 5. R3 端口镜像配置命令



```
2025-12-15 21:08.12 /home/mobaxterm telnet 172.16.17.203
Connected to 172.16.17.203.
Entering character mode
Escape character is '^]'.

Login authentication

Username:user
Password:
-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.17.3
Time      : 2025-12-15 13:04:04+00:00
-----
<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]observe-port interface GigabitEthernet 0/0/2
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]mirror to observe-port both
Warning: Get packets will be sent to observe port
[R3-GigabitEthernet0/0/0]quit
[R3]quit
<R3>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 8. 在 R3 上配置端口镜像

1.3.3 第三步：抓包并配置 OSPF 与认证 (任务 a, b)

操作描述：在 PC3 上启动 Wireshark，然后按照 R1、R2、R3、R4 的顺序，依次在各路由器上启动 OSPF 进程，宣告各自的直连网段，并配置密码为‘123456’的区域认证。这个过程允许我们观察 OSPF 网络逐步建立的过程。

```
system-view
ospf 1 router-id 1.1.1.1
area 0
authentication-mode simple plain 123456
network 192.168.12.0 0.0.0.255
network 192.168.13.0 0.0.0.255
quit
quit
```

Listing 6. R1 OSPF 与认证配置命令

```
2025-12-15 21:01:42 /home/mobaxterm telnet 172.16.15.203
Connected to 172.16.15.203.
Entering character mode
Escape character is '^J'.

Login authentication

Username:user
Password:
-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.15.3
Time : 2025-12-15 12:58:32+00:00
-----
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]ospf 1 router-id 1.1.1.1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]authentication-mode simple plain 123456
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R1-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]network 192.168.13.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]quit
<R1>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 9. R1 OSPF 与认证配置过程

```
system-view
ospf 1 router-id 2.2.2.2
    area 0
        authentication-mode simple plain 123456
        network 192.168.12.0 0.0.0.255
        network 192.168.24.0 0.0.0.255
    quit
quit
```

Listing 7. R2 OSPF 与认证配置命令

```
2025-12-15 21:28:15 /home/mobaxterm telnet 172.16.15.204
Connected to 172.16.15.204.
Entering character mode
Escape character is '^J'.

Login authentication

Username:user
Password:
-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.15.3
Time : 2025-12-15 21:45:40+00:00
-----
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]ospf 1 router-id 2.2.2.2
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]authentication-mode simple plain 123456
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R2-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]network 192.168.24.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]quit
[R2-ospf-1]quit
[R2]quit
<R2>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 10. R2 OSPF 与认证配置过程

Wireshark 抓包分析:

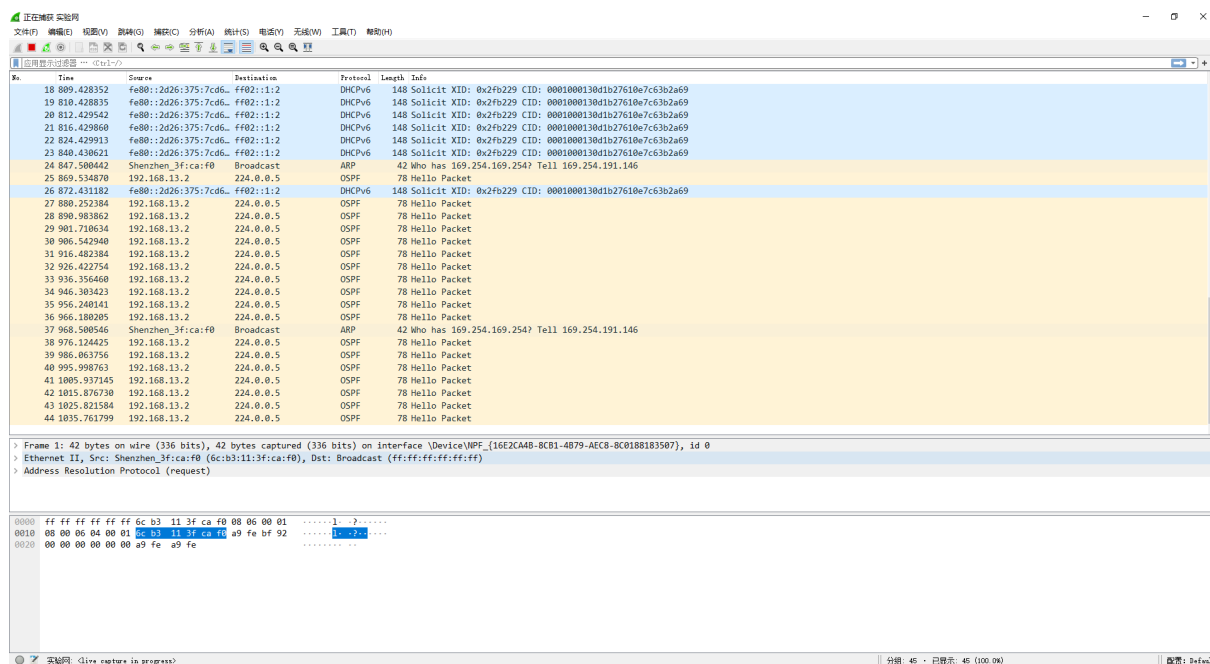


图 11. 仅配置 R1 和 R2 后, Wireshark 捕获到两者互相发送的 Hello 报文

分析: 仅配置完核心路由器 R1 和 R2 后, 镜像端口已能捕获到它们之间周期性发送的 Hello 报文, 但此时网络尚未形成完整邻接关系, 没有路由学习。

```
system-view
ospf 1 router-id 3.3.3.3
area 0
authentication-mode simple plain 123456
network 192.168.1.0 0.0.0.255
network 192.168.13.0 0.0.0.255
network 192.168.34.0 0.0.0.255
quit
quit
```

Listing 8. R3 OSPF 与认证配置命令

```
2025-12-15 20:45:49 /home/mobaxterm telnet 172.16.17.203
Connected to 172.16.17.203.
Entering character mode.
Escape character is '^J'.

Login authentication

Username:user
Password:

-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.17.3
Time      : 2025-12-15 13:12:52+00:00
-----

<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]ospf 1 router-id 3.3.3.3
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]authentication-mode simple plain 123456
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R3-ospf-1-area-0.0.0.0]network 192.168.1.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]network 192.168.13.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]network 192.168.34.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]quit
[R3-ospf-1]quit
[R3]quit
<R3>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 12. R3 OSPF 与认证配置过程

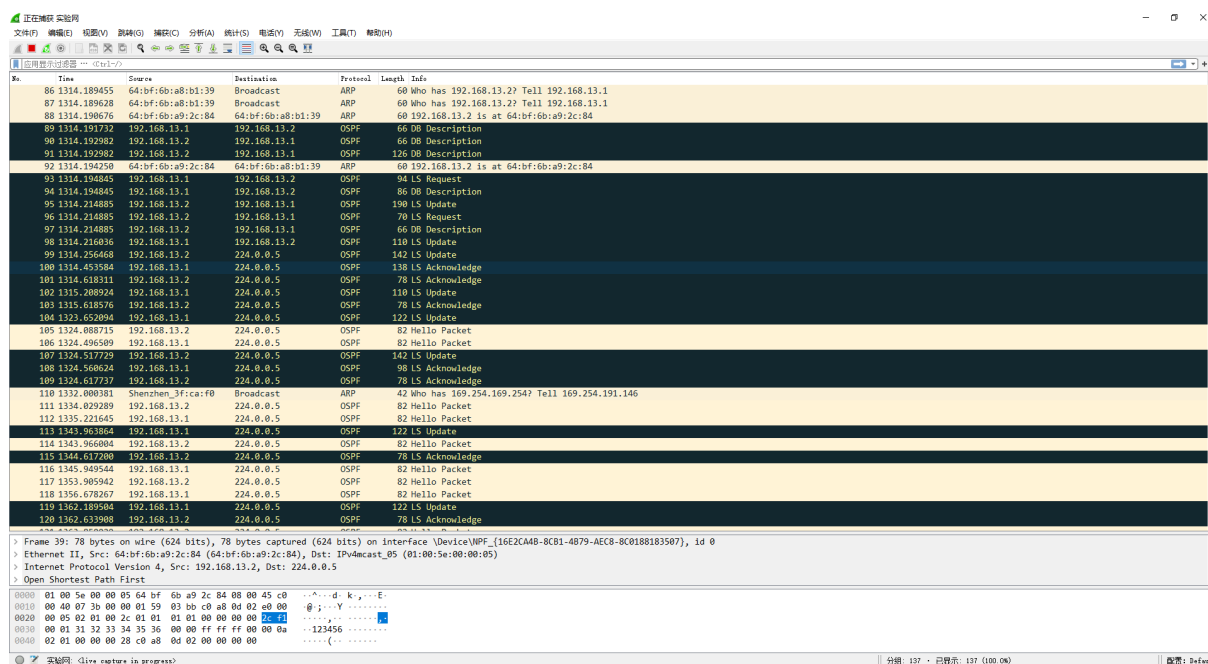


图 13. 配置 R3 后，捕获到 R1 与 R3 建立邻居关系的全过程报文

分析: 当边缘路由器 R3 配置完成后，Wireshark 立刻捕获到 R1 与 R3 之间密集的报文交换，包括 Hello, DB Description, LS Request, LS Update 及 LS Acknowledge，这清晰地展示了 OSPF 邻居从无到有的建立过程。下面是五种关键报文的详细分析：

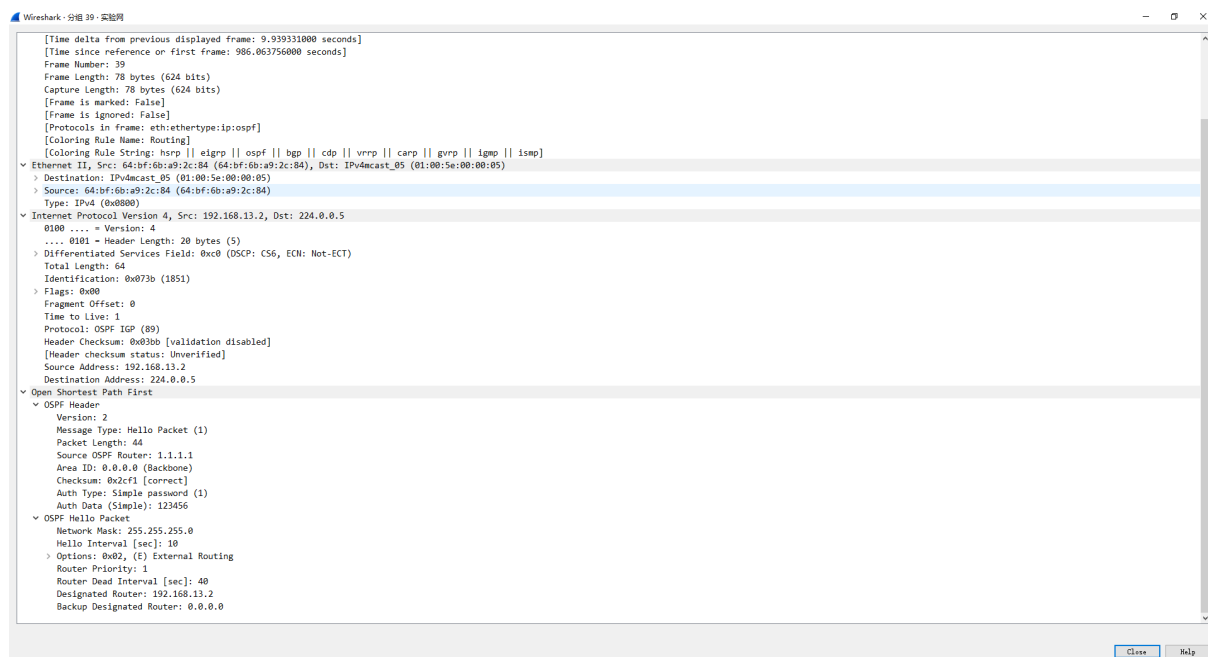


图 14. Hello 报文 (Type 1) 详情

Hello 报文分析: 用于发现、建立和维护邻居关系。从此报文详情中可以看到认证类型为 ‘Simple password’，密码为 ‘123456’，证明了区域认证配置成功。

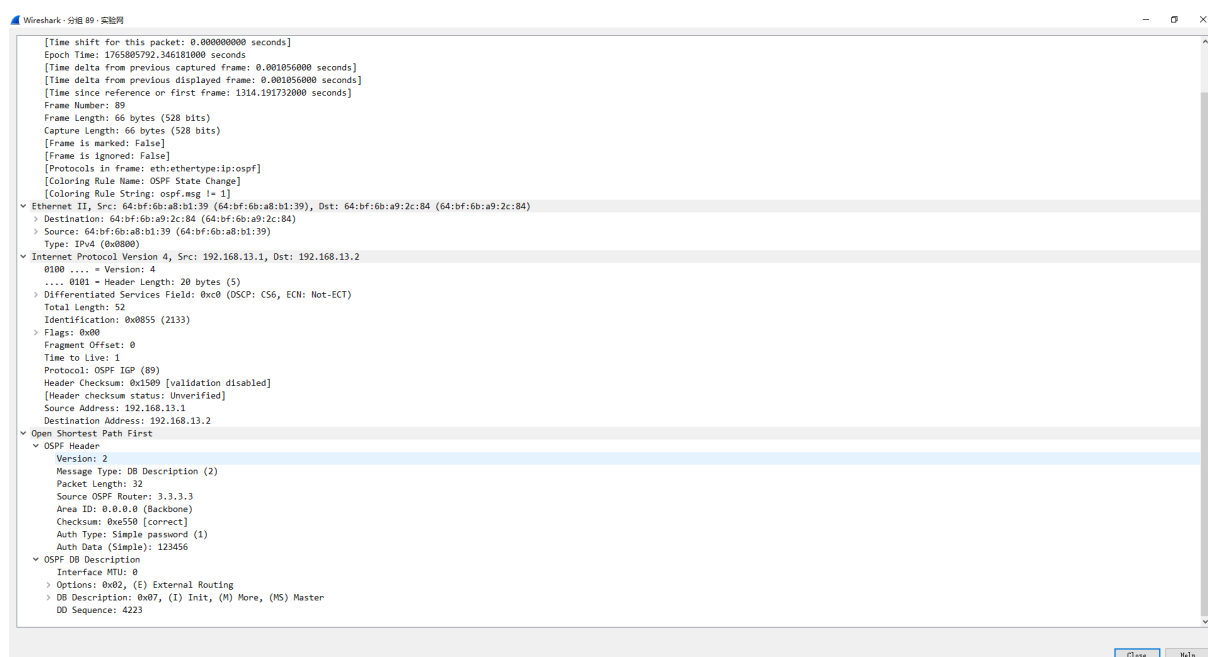


图 15. DB Description 报文 (Type 2) 详情

DD 报文分析: 用于同步 LSDB 摘要，是数据库同步的开始。

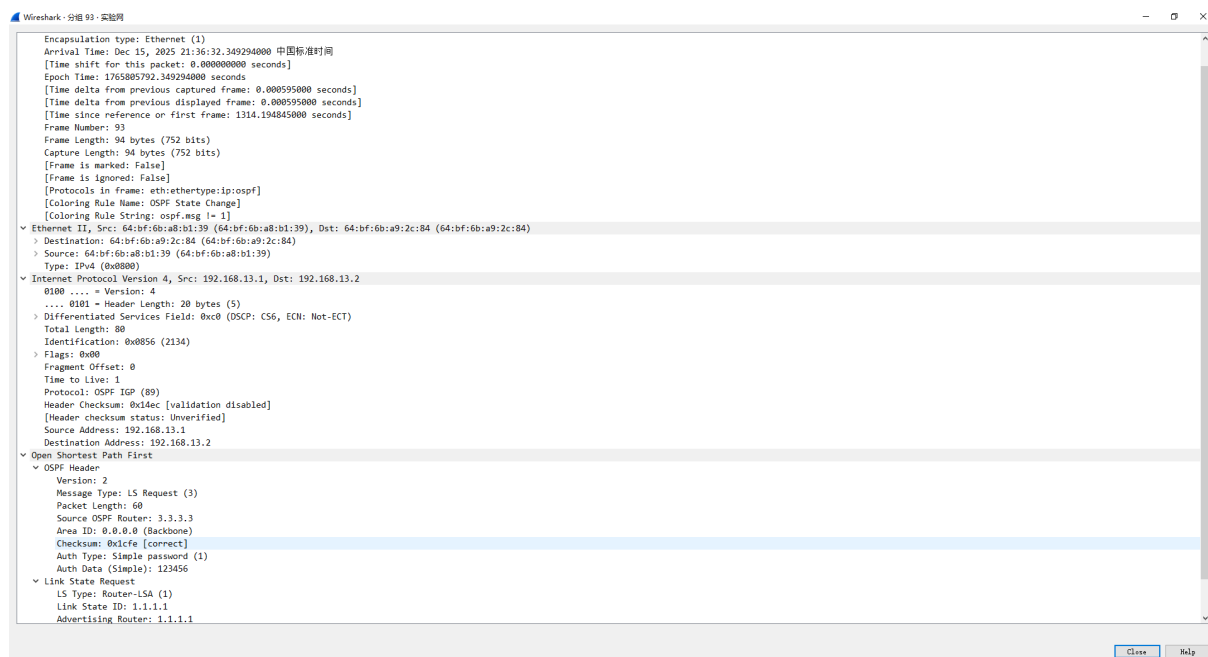


图 16. LS Request 报文 (Type 3) 详情

LSR 报文分析: 用于向邻居请求缺失的 LSA，实现了按需更新。

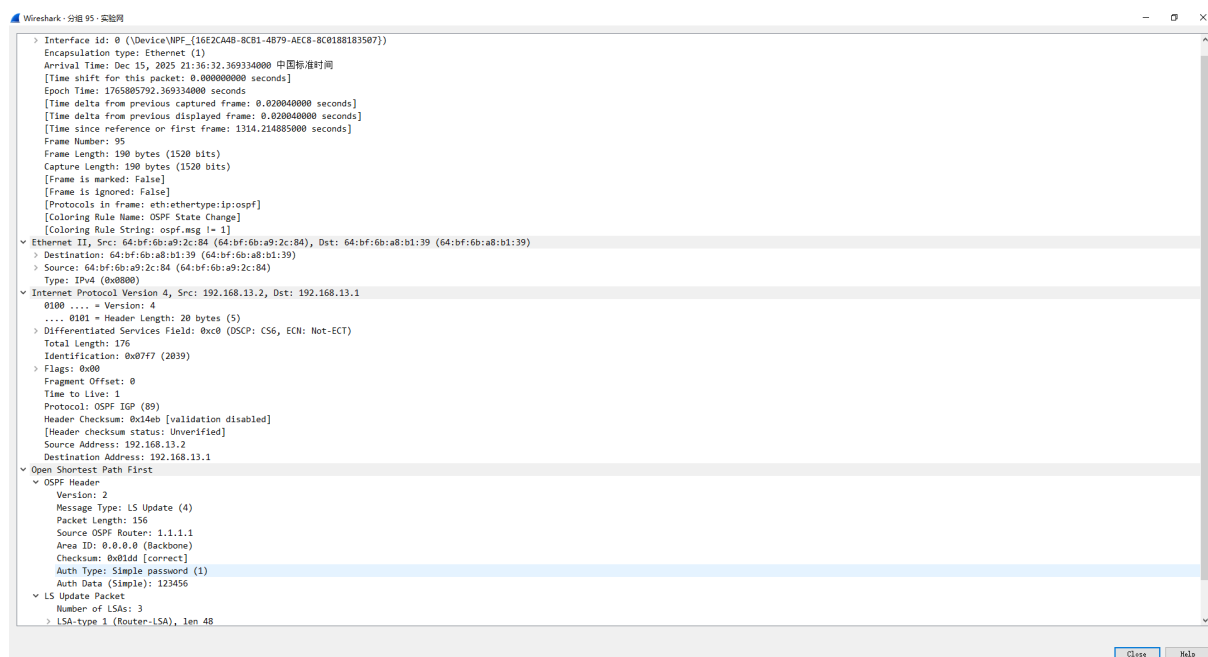


图 17. LS Update 报文 (Type 4) 详情

LSU 报文分析: OSPF 信息交换的核心，承载了具体的 LSA 拓扑信息。

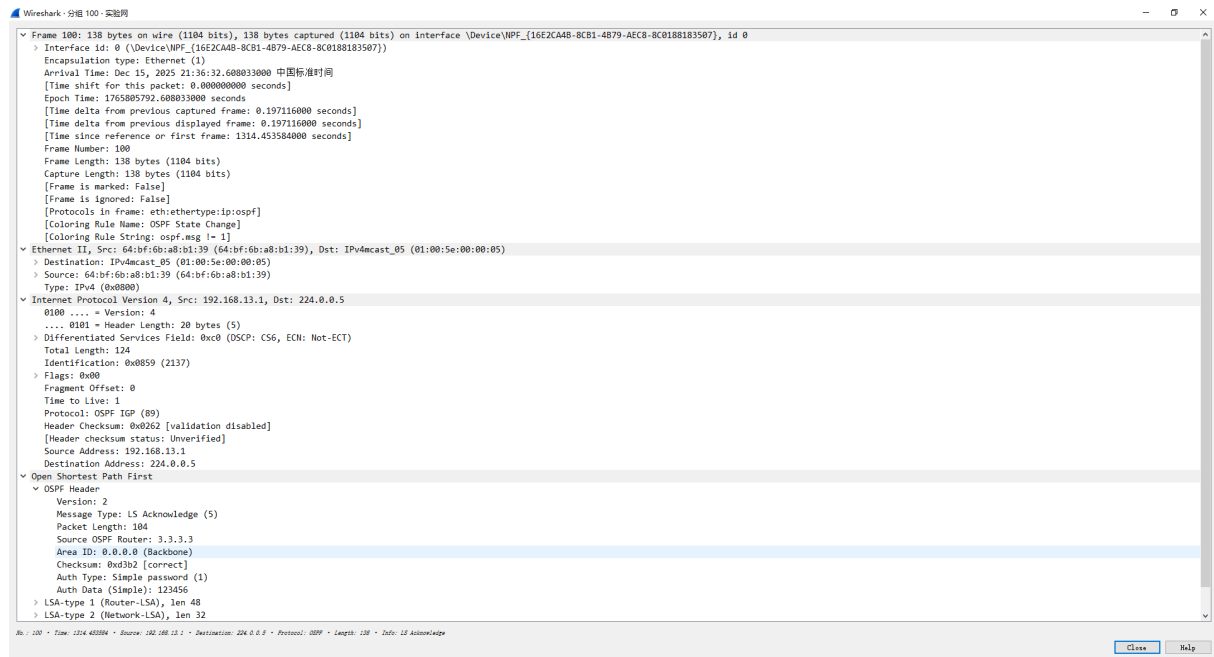


图 18. LS Acknowledge 报文 (Type 5) 详情

LSAck 报文分析: 用于确认收到的 LSU，保证了信息交换的可靠性。

```
system-view
ospf 1 router-id 4.4.4.4
area 0
authentication-mode simple plain 123456
network 192.168.2.0 0.0.0.255
network 192.168.24.0 0.0.0.255
network 192.168.34.0 0.0.0.255
quit
quit
```

Listing 9. R4 OSPF 与认证配置命令


```
2025-12-15 21:44:53 /home/mobaxterm telnet 172.16.17.204
Connected to 172.16.17.204.
Entering character mode
Escape character is '^'.

Login authentication

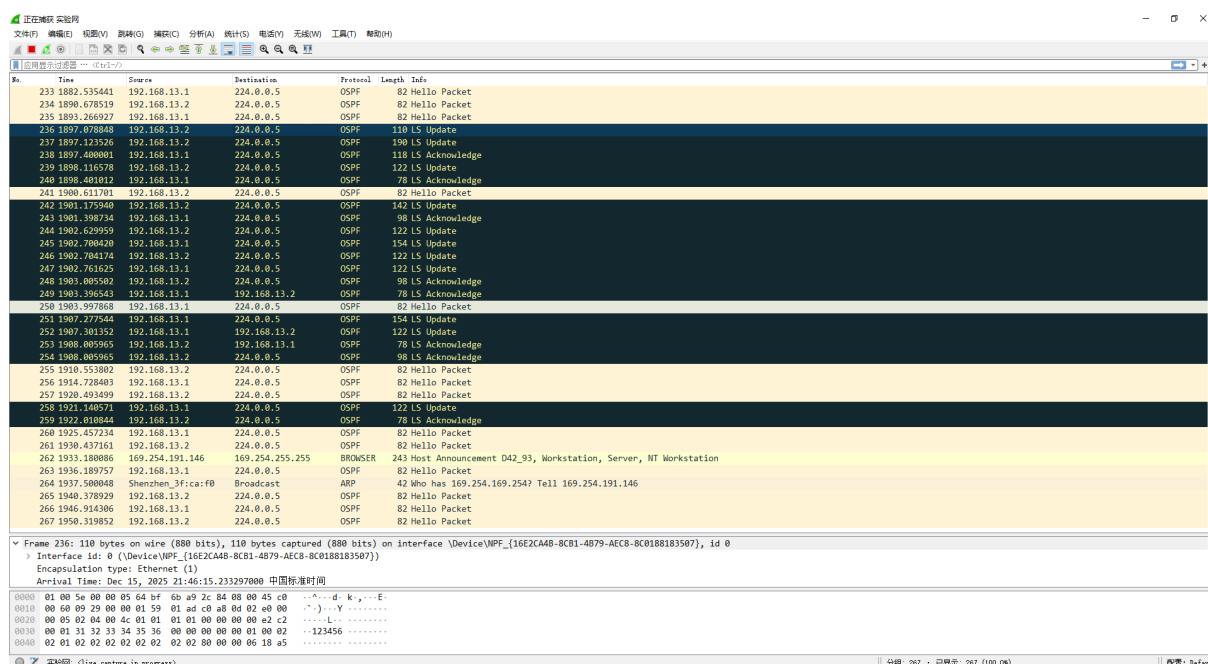
Username:user
Password:

-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.17.3
Time      : 2025-12-15 22:13:12+00:00
-----

<R4>system-view
Enter system view, return user view with Ctrl+Z.
[R4]ospf 1 router-id 4.4.4.4
[R4-ospf-1]area 0
[R4-ospf-1-area-0.0.0.0]authentication-mode simple plain 123456
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R4-ospf-1-area-0.0.0.0]network 192.168.2.0 0.0.0.255
[R4-ospf-1-area-0.0.0.0]network 192.168.24.0 0.0.0.255
[R4-ospf-1-area-0.0.0.0]network 192.168.34.0 0.0.0.255
[R4-ospf-1-area-0.0.0.0]quit
[R4-ospf-1]quit
[R4]quit
<R4>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 19. R4 OSPF 与认证配置过程



No.	Time	Source	Destination	Protocol	Length	Info
233	1882.535441	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
234	1890.678519	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
235	1893.266927	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
236	1897.078848	192.168.13.2	224.0.0.5	OSPF	110	LS Update
237	1897.123526	192.168.13.2	224.0.0.5	OSPF	198	LS Update
238	1897.480801	192.168.13.1	224.0.0.5	OSPF	118	LS Acknowledge
239	1898.116578	192.168.13.2	224.0.0.5	OSPF	122	LS Update
240	1898.481812	192.168.13.1	224.0.0.5	OSPF	78	LS Acknowledge
241	1900.611781	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
242	1901.175980	192.168.13.2	224.0.0.5	OSPF	142	Update
243	1901.388784	192.168.13.1	224.0.0.5	OSPF	98	LS Acknowledge
244	1902.629959	192.168.13.2	224.0.0.5	OSPF	122	LS Update
245	1902.780420	192.168.13.1	224.0.0.5	OSPF	154	LS Update
246	1902.784174	192.168.13.2	224.0.0.5	OSPF	122	LS Update
247	1902.761625	192.168.13.1	224.0.0.5	OSPF	122	LS Update
248	1903.085582	192.168.13.2	224.0.0.5	OSPF	98	LS Acknowledge
249	1903.396543	192.168.13.1	192.168.13.2	OSPF	78	LS Acknowledge
250	1903.997868	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
251	1907.277544	192.168.13.1	224.0.0.5	OSPF	154	LS Update
252	1907.381352	192.168.13.1	192.168.13.2	OSPF	122	LS Update
253	1908.085965	192.168.13.2	192.168.13.1	OSPF	78	LS Acknowledge
254	1908.085965	192.168.13.2	224.0.0.5	OSPF	98	LS Acknowledge
255	1910.553802	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
256	1914.728483	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
257	1920.493489	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
258	1921.148571	192.168.13.1	224.0.0.5	OSPF	122	LS Update
259	1922.018844	192.168.13.2	224.0.0.5	OSPF	78	LS Acknowledge
260	1925.457234	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
261	1930.437161	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
262	1933.188086	169.254.191.146	169.254.255.255	BROADCAST	243	Host Announcement D42_93, Workstation, Server, IIT Workstation
263	1936.189757	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
264	1937.508048	Shenzhen_3f:ca:f0	Broadcast	ARP	42	Who has 169.254.169.254? Tell 169.254.191.146
265	1940.378929	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
266	1946.914386	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
267	1950.319852	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet

图 20. 配置 R4 后，捕获到网络中新增的 LSU 和 LSAck 报文

分析：当网络已稳定，最后加入 R4 时，已有路由器（如 R3）主要向新邻居 R4 发送已知的拓扑信息（LS Update），并接收 R4 的确认（LS Acknowledge），体现了 OSPF 的增量更新机制。

1.3.4 第四步：验证 OSPF 状态与连通性（拔线前）

操作描述：检查所有路由器 OSPF 邻居和路由表状态，并进行端到端连通性测试。

```

[R1]display ospf peer

      OSPF Process 1 with Router ID 1.1.1.1
        Neighbors

Area 0.0.0.0 interface 192.168.13.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 3.3.3.3      Address: 192.168.13.1
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.13.2  BDR: 192.168.13.1  MTU: 0
  Dead timer due in 35 sec
  Retrans timer interval: 5
  Neighbor is up for 00:21:44
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.12.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 2.2.2.2      Address: 192.168.12.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.12.1  BDR: 192.168.12.2  MTU: 0
  Dead timer due in 32 sec
  Retrans timer interval: 6
  Neighbor is up for 00:27:14
  Authentication Sequence: [ 0 ]

[R1]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
      Destinations : 17          Routes : 18

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
127.0.0.0/8         Direct   0    0              D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct   0    0              D    127.0.0.1        InLoopBack0
127.255.255.255/32  Direct   0    0              D    127.0.0.1        InLoopBack0
172.16.0.0/16       Direct   0    0              D    172.16.15.203    GigabitEthernet0/0/8
172.16.15.203/32    Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/8
172.16.255.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/8
192.168.1.0/24       OSPF     10    2              D    192.168.13.1     GigabitEthernet0/0/0
192.168.2.0/24       OSPF     10    3              D    192.168.12.2     GigabitEthernet0/0/1
                   OSPF     10    3              D    192.168.13.1     GigabitEthernet0/0/0
192.168.12.0/24     Direct   0    0              D    192.168.12.1     GigabitEthernet0/0/1
192.168.12.1/32     Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/1
192.168.12.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/1
192.168.13.0/24     Direct   0    0              D    192.168.13.2     GigabitEthernet0/0/0
192.168.13.2/32     Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/0
192.168.13.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/0
192.168.24.0/24     OSPF     10    2              D    192.168.12.2     GigabitEthernet0/0/1
192.168.34.0/24     OSPF     10    2              D    192.168.13.1     GigabitEthernet0/0/0
255.255.255.255/32  Direct   0    0              D    127.0.0.1        InLoopBack0

```

图 21. R1 的 OSPF 邻居与路由表（拔线前）

```
[R2]display ospf peer

      OSPF Process 1 with Router ID 2.2.2.2
        Neighbors

Area 0.0.0.0 interface 192.168.12.2(GigabitEthernet0/0/7)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.12.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.12.1 BDR: 192.168.12.2 MTU: 0
Dead timer due in 40 sec
Retrans timer interval: 0
Neighbor is up for 00:27:59
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.24.1(GigabitEthernet0/0/5)'s neighbors
Router ID: 4.4.4.4      Address: 192.168.24.2
State: Full Mode:Nbr is Master Priority: 1
DR: 192.168.24.1 BDR: 192.168.24.2 MTU: 0
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00:12:47
Authentication Sequence: [ 0 ]

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 17          Routes : 18

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
127.0.0.0/8         Direct   0    0              D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct   0    0              D    127.0.0.1        InLoopBack0
127.255.255.255/32  Direct   0    0              D    127.0.0.1        InLoopBack0
172.16.0.0/16       Direct   0    0              D    172.16.15.204    GigabitEthernet0/0/9
172.16.15.204/32    Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/9
192.168.1.0/24       OSPF     10   3              D    192.168.12.1     GigabitEthernet0/0/7
                   OSPF     10   3              D    192.168.24.2     GigabitEthernet0/0/5
192.168.2.0/24       OSPF     10   2              D    192.168.24.2     GigabitEthernet0/0/5
192.168.12.0/24     Direct   0    0              D    192.168.12.2     GigabitEthernet0/0/7
192.168.12.2/32     Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/7
192.168.12.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/7
192.168.13.0/24     OSPF     10   2              D    192.168.12.1     GigabitEthernet0/0/7
192.168.24.0/24     Direct   0    0              D    192.168.24.1     GigabitEthernet0/0/5
192.168.24.1/32     Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/5
192.168.24.255/32   Direct   0    0              D    127.0.0.1        GigabitEthernet0/0/5
192.168.34.0/24     OSPF     10   2              D    192.168.24.2     GigabitEthernet0/0/5
255.255.255.255/32  Direct   0    0              D    127.0.0.1        InLoopBack0
```

图 22. R2 的 OSPF 邻居与路由表（拔线前）

```
[R3]display ospf peer

      OSPF Process 1 with Router ID 3.3.3.3
        Neighbors

Area 0.0.0.0 interface 192.168.13.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.13.2
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 192.168.13.2  BDR: 192.168.13.1  MTU: 0
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00:17:05
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.34.1(GigabitEthernet0/0/3)'s neighbors
Router ID: 4.4.4.4      Address: 192.168.34.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.34.1  BDR: 192.168.34.2  MTU: 0
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00:07:17
  Authentication Sequence: [ 0 ]
```

图 23. R3 的 OSPF 邻居状态（拔线前）

```
[R3]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
  Destinations : 19          Routes : 19

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
127.0.0.0/8         Direct   0    0        D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1         InLoopBack0
127.255.255.255/32   Direct   0    0        D    127.0.0.1         InLoopBack0
172.16.0.0/16        Direct   0    0        D    172.16.17.203     GigabitEthernet0/0/8
172.16.17.203/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/8
172.16.255.255/32    Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/8
192.168.1.0/24        Direct   0    0        D    192.168.1.1       GigabitEthernet0/0/1
192.168.1.1/32        Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/1
192.168.1.255/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/1
192.168.2.0/24        OSPF     10    2        D    192.168.34.2       GigabitEthernet0/0/3
192.168.12.0/24       OSPF     10    2        D    192.168.13.2       GigabitEthernet0/0/0
192.168.13.0/24       Direct   0    0        D    192.168.13.1       GigabitEthernet0/0/0
192.168.13.1/32       Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/0
192.168.13.255/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/0
192.168.24.0/24       OSPF     10    2        D    192.168.34.2       GigabitEthernet0/0/3
192.168.34.0/24       Direct   0    0        D    192.168.34.1       GigabitEthernet0/0/3
192.168.34.1/32       Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/3
192.168.34.255/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/3
255.255.255.255/32    Direct   0    0        D    127.0.0.1         InLoopBack0
```

图 24. R3 的路由表（拔线前）

```
[R4]display ospf peer

      OSPF Process 1 with Router ID 4.4.4.4
        Neighbors

Area 0.0.0.0 interface 192.168.24.2(GigabitEthernet0/0/5)'s neighbors
Router ID: 2.2.2.2      Address: 192.168.24.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.24.1 BDR: 192.168.24.2 MTU: 0
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00:09:06
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.34.2(GigabitEthernet0/0/8)'s neighbors
Router ID: 3.3.3.3      Address: 192.168.34.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.34.1 BDR: 192.168.34.2 MTU: 0
Dead timer due in 28 sec
Retrans timer interval: 5
Neighbor is up for 00:09:01
Authentication Sequence: [ 0 ]

[R4]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 19          Routes : 19

Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
127.0.0.0/8         Direct   0    0        D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1        InLoopBack0
127.255.255.255/32  Direct   0    0        D    127.0.0.1        InLoopBack0
172.16.0.0/16       Direct   0    0        D    172.16.17.204    GigabitEthernet0/0/9
172.16.17.204/32    Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/9
192.168.1.0/24       OSPF     10    2        D    192.168.34.1     GigabitEthernet0/0/8
192.168.2.0/24       Direct   0    0        D    192.168.2.1      GigabitEthernet0/0/7
192.168.2.1/32       Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/7
192.168.2.255/32     Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/7
192.168.12.0/24      OSPF     10    2        D    192.168.24.1     GigabitEthernet0/0/5
192.168.13.0/24      OSPF     10    2        D    192.168.34.1     GigabitEthernet0/0/8
192.168.24.0/24      Direct   0    0        D    192.168.24.2     GigabitEthernet0/0/5
192.168.24.2/32      Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/5
192.168.24.255/32    Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/5
192.168.34.0/24      Direct   0    0        D    192.168.34.2     GigabitEthernet0/0/8
192.168.34.2/32      Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/8
192.168.34.255/32    Direct   0    0        D    127.0.0.1        GigabitEthernet0/0/8
255.255.255.255/32   Direct   0    0        D    127.0.0.1        InLoopBack0
```

图 25. R4 的 OSPF 邻居与路由表（拔线前）

状态分析: 从各路由器 ‘display ospf peer’ 的输出可以看到，每台路由器都与其物理直连的邻居成功建立起了 ‘Full’ 状态的邻接关系。‘display ip routing-table’ 的输出则显示，每台路由器都通过 OSPF 学习到了全网所有非直连网段的路由信息，证明了路由已自动收敛。

```
C:\Users\Administrator>ping 192.168.2.10

正在 Ping 192.168.2.10 具有 32 字节的数据:
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=126

192.168.2.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 0ms, 平均 = 0ms
```

图 26. 连通性测试: PC1 ping PC2 成功

```
C:\Users\Administrator>ping 192.168.1.10

正在 Ping 192.168.1.10 具有 32 字节的数据:
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=126
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=126

192.168.1.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 0ms, 平均 = 0ms
```

图 27. 连通性测试: PC2 ping PC1 成功

连通性测试结果: 可以看出, 此时 PC1 与 PC2 之间已经可以相互 ping 通。

1.3.5 第五步: 故障模拟与路径变化 (任务 c)

操作描述: 首先通过 tracert 命令查看当前数据路径, 然后拔掉 R3 与 R4 之间的冗余链路, 观察 OSPF 的自愈合能力。

```
C:\Users\Administrator>tracert 192.168.2.10

通过最多 30 个跃点跟踪到 192.168.2.10 的路由

  1      1 ms      <1 毫秒      <1 毫秒  192.168.1.1
  2      <1 毫秒    <1 毫秒      <1 毫秒  192.168.34.2
  3      <1 毫秒    <1 毫秒      <1 毫秒  192.168.2.10

跟踪完成。
```

图 28. 故障前：PC1 到 PC2 的数据路径

```
C:\Users\Administrator>tracert 192.168.1.10

通过最多 30 个跃点跟踪到 192.168.1.10 的路由

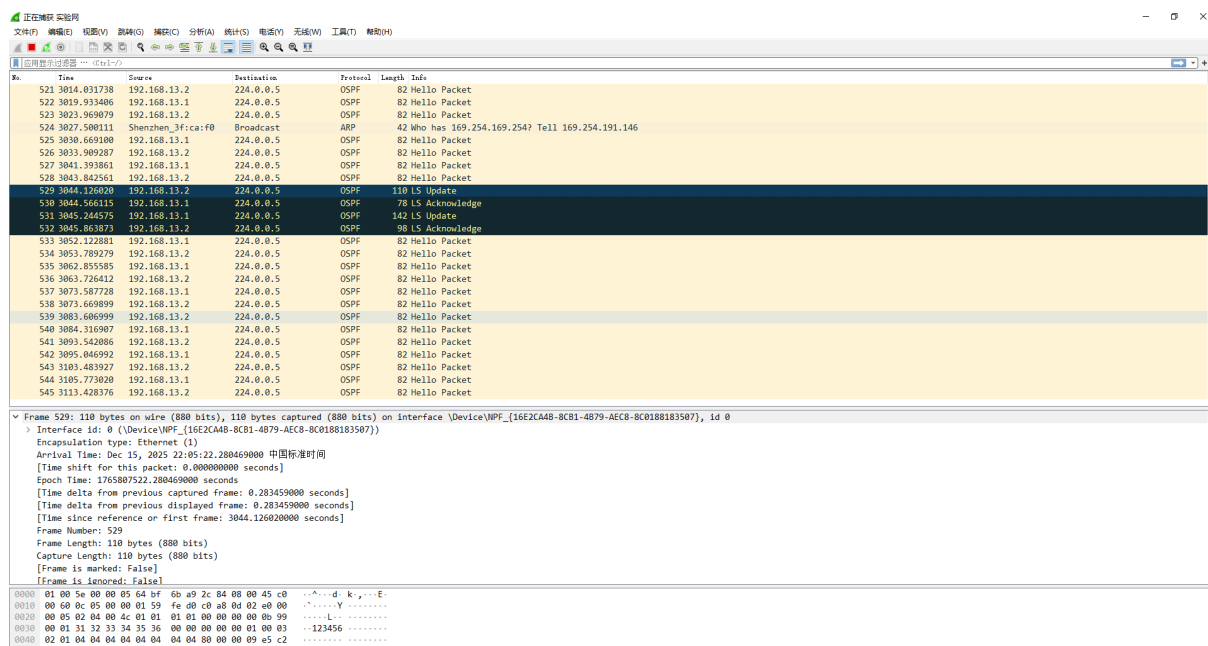
  1      <1 毫秒    <1 毫秒      <1 毫秒  192.168.2.1
  2      1 ms      <1 毫秒      <1 毫秒  192.168.34.1
  3      <1 毫秒    <1 毫秒      <1 毫秒  192.168.1.10

跟踪完成。
```

图 29. 故障前：PC2 到 PC1 的数据路径

路径分析：拔线前，由于 R3-R4 链路（192.168.34.0/24）存在，OSPF 计算出的最短路径是通过 R3 直连 R4。tracert 结果显示路径为 PC1 -> R3 -> R4 -> PC2（PC2 到 PC1 则相反），共 3 跳。

操作：拔掉 R3 与 R4 之间的冗余链路网线。等待约 40 秒后，再次进行验证。



No.	Time	Source	Destination	Protocol	Length	Info
521	3014.831738	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
522	3019.933406	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
523	3023.969079	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
524	3027.580111	Shenzhen:3f:ca:f0	Broadcast	ARP	42	Who has 169.254.169.254? Tell 169.254.191.146
525	3030.669100	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
526	3033.909287	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
527	3041.393861	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
528	3043.842561	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
529	3044.126020	192.168.13.2	224.0.0.5	OSPF	110	LS Update
530	3044.566115	192.168.13.1	224.0.0.5	OSPF	78	LS Acknowledge
531	3045.244575	192.168.13.1	224.0.0.5	OSPF	142	LS Update
532	3045.863873	192.168.13.2	224.0.0.5	OSPF	98	LS Acknowledge
533	3052.122881	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
534	3053.709279	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
535	3062.855585	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
536	3063.726412	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
537	3073.587728	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
538	3073.669899	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
539	3083.686999	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
540	3084.316907	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
541	3093.542086	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
542	3095.046992	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
543	3103.483927	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
544	3105.773020	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
545	3113.428376	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet

Frame 529: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface \Device\NPF_{16E2CA48-8CB1-4B79-AEC8-8C0188183507}, Id 0
Interface Id: 0 (\Device\NPF_{16E2CA48-8CB1-4B79-AEC8-8C0188183507})
Encapsulation type: Ethernet (1)
Arrival Time: Dec 15, 2025 22:05:22.280469000 中国标准时间
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1765807522.280469000 seconds
[Time delta from previous captured frame: 0.283459000 seconds]
[Time delta from previous displayed frame: 0.283459000 seconds]
[Time since reference or first frame: 3044.126020000 seconds]
Frame Number: 529
Frame Length: 110 bytes (880 bits)
Capture Length: 110 bytes (880 bits)
[Frame is marked: False]
[Frame is ignored: False]

0000 01 00 5e 00 00 05 64 bf 6b a9 2c 84 08 00 45 c0 ...^...d.k.....E:
0010 00 60 8c 05 00 00 01 59 fe 00 c0 a8 04 02 c0 00Y.....
0020 00 05 02 04 00 4c 01 01 01 01 00 00 00 00 99L.....
0030 00 01 31 32 33 34 35 36 00 00 00 00 00 01 00 03 ...123456.....
0040 02 01 04 04 04 04 04 04 04 00 00 00 09 e5 c2</p></div>
<div data-bbox="234 384 762 401" data-label="Caption">
图 30. 拔线后，Wireshark 再次捕获到宣告拓扑变化的 LSU 报文
</div>
<div data-bbox="114 433 882 477" data-label="Text">
<p>分析:拔掉网线后，R3 与 R4 邻居关系超时中断，它们立刻生成新的 LSA 宣告拓扑变化，并通过 LSU 泛洪至全网。</p>
</div>
<div data-bbox="485 935 512 952" data-label="Page-Footer">24</div>


```
[R1]display ospf peer

      OSPF Process 1 with Router ID 1.1.1.1
        Neighbors

Area 0.0.0.0 interface 192.168.13.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 3.3.3.3      Address: 192.168.13.1
State: ExStart Mode:Nbr is Slave Priority: 1
DR: None BDR: None MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 0
Neighbor is up for 00:00:00
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.12.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 2.2.2.2      Address: 192.168.12.2
State: Full Mode:Nbr is Master Priority: 1
DR: 192.168.12.1 BDR: 192.168.12.2 MTU: 0
Dead timer due in 30 sec
Retrans timer interval: 6
Neighbor is up for 00:41:55
Authentication Sequence: [ 0 ]

[R1]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
  Destinations : 16      Routes : 16

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
127.0.0.0/8         Direct   0    0        D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1         InLoopBack0
127.255.255.255/32   Direct   0    0        D    127.0.0.1         InLoopBack0
172.16.0.0/16       Direct   0    0        D    172.16.15.203     GigabitEthernet0/0/8
172.16.15.203/32    Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/8
172.16.255.255/32   Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/8
192.168.1.0/24       OSPF     10    2        D    192.168.13.1      GigabitEthernet0/0/0
192.168.2.0/24       OSPF     10    3        D    192.168.12.2      GigabitEthernet0/0/1
192.168.12.0/24      Direct   0    0        D    192.168.12.1      GigabitEthernet0/0/1
192.168.12.1/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/1
192.168.12.255/32   Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/1
192.168.13.0/24     Direct   0    0        D    192.168.13.2      GigabitEthernet0/0/0
192.168.13.2/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/0
192.168.13.255/32   Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/0
192.168.24.0/24     OSPF     10    2        D    192.168.12.2      GigabitEthernet0/0/1
255.255.255.255/32   Direct   0    0        D    127.0.0.1         InLoopBack0
```

图 31. 拔线后 R1 的邻居与路由表

```
[R2]display ospf peer

      OSPF Process 1 with Router ID 2.2.2.2
        Neighbors

Area 0.0.0.0 interface 192.168.12.2(GigabitEthernet0/0/7)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.12.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.12.1 BDR: 192.168.12.2 MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:42:57
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.24.1(GigabitEthernet0/0/5)'s neighbors
Router ID: 4.4.4.4      Address: 192.168.24.2
State: Full Mode:Nbr is Master Priority: 1
DR: 192.168.24.1 BDR: 192.168.24.2 MTU: 0
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00:27:45
Authentication Sequence: [ 0 ]

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
-----
127.0.0.0/8         Direct   0    0              D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0              D    127.0.0.1         InLoopBack0
127.255.255.255/32  Direct   0    0              D    127.0.0.1         InLoopBack0
172.16.0.0/16       Direct   0    0              D    172.16.15.204     GigabitEthernet0/0/9
172.16.15.204/32    Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/9
192.168.1.0/24       OSPF     10    3              D    192.168.12.1       GigabitEthernet0/0/7
192.168.2.0/24       OSPF     10    2              D    192.168.24.2       GigabitEthernet0/0/5
192.168.12.0/24     Direct   0    0              D    192.168.12.2       GigabitEthernet0/0/7
192.168.12.2/32     Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/7
192.168.12.255/32   Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/7
192.168.13.0/24     OSPF     10    2              D    192.168.12.1       GigabitEthernet0/0/7
192.168.24.0/24     Direct   0    0              D    192.168.24.1       GigabitEthernet0/0/5
192.168.24.1/32     Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/5
192.168.24.255/32   Direct   0    0              D    127.0.0.1         GigabitEthernet0/0/5
255.255.255.255/32  Direct   0    0              D    127.0.0.1         InLoopBack0
```

图 32. 拔线后 R2 的邻居与路由表

```

[R3]display ospf peer

      OSPF Process 1 with Router ID 3.3.3.3
        Neighbors

Area 0.0.0.0 interface 192.168.13.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.13.2
State: Full  Mode:Nbr is Slave Priority: 1
DR: 192.168.13.2  BDR: 192.168.13.1  MTU: 0
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00:30:50
Authentication Sequence: [ 0 ]

[R3]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
      Destinations : 16          Routes : 16

Destination/Mask    Proto    Pre  Cost      Flags NextHop         Interface
127.0.0.0/8         Direct  0    0          D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct  0    0          D    127.0.0.1        InLoopBack0
127.255.255.255/32   Direct  0    0          D    127.0.0.1        InLoopBack0
172.16.0.0/16        Direct  0    0          D    172.16.17.203    GigabitEthernet0/0/8
172.16.17.203/32     Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/8
172.16.255.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/8
192.168.1.0/24       Direct  0    0          D    192.168.1.1      GigabitEthernet0/0/1
192.168.1.1/32       Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/1
192.168.1.255/32     Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/1
192.168.2.0/24       OSPF    10    4          D    192.168.13.2     GigabitEthernet0/0/0
192.168.12.0/24      OSPF    10    2          D    192.168.13.2     GigabitEthernet0/0/0
192.168.13.0/24      Direct  0    0          D    192.168.13.1     GigabitEthernet0/0/0
192.168.13.1/32      Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/0
192.168.13.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet0/0/0
192.168.24.0/24      OSPF    10    3          D    192.168.13.2     GigabitEthernet0/0/0
255.255.255.255/32    Direct  0    0          D    127.0.0.1        InLoopBack0

```

图 33. 拔线后 R3 的邻居与路由表

```
[R4]display ospf peer

      OSPF Process 1 with Router ID 4.4.4.4
        Neighbors

Area 0.0.0.0 interface 192.168.24.2(GigabitEthernet0/0/5)'s neighbors
Router ID: 2.2.2.2      Address: 192.168.24.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.24.1 BDR: 192.168.24.2 MTU: 0
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00:21:53
Authentication Sequence: [ 0 ]

[R4]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
127.0.0.0/8         Direct   0    0        D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1         InLoopBack0
127.255.255.255/32  Direct   0    0        D    127.0.0.1         InLoopBack0
172.16.0.0/16       Direct   0    0        D    172.16.17.204     GigabitEthernet0/0/9
172.16.17.204/32    Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/9
192.168.1.0/24      OSPF     10    4        D    192.168.24.1      GigabitEthernet0/0/5
192.168.2.0/24      Direct   0    0        D    192.168.2.1       GigabitEthernet0/0/7
192.168.2.1/32      Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/7
192.168.2.255/32    Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/7
192.168.12.0/24     OSPF     10    2        D    192.168.24.1      GigabitEthernet0/0/5
192.168.13.0/24     OSPF     10    3        D    192.168.24.1      GigabitEthernet0/0/5
192.168.24.0/24     Direct   0    0        D    192.168.24.2      GigabitEthernet0/0/5
192.168.24.2/32     Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/5
192.168.24.255/32   Direct   0    0        D    127.0.0.1         GigabitEthernet0/0/5
255.255.255.255/32  Direct   0    0        D    127.0.0.1         InLoopBack0
```

图 34. 拔线后 R4 的邻居与路由表

状态分析: 拔线后, R3 与 R4 的邻居关系消失。R3 路由表中去往 PC2(192.168.2.0/24) 的路径依然存在, 但 Cost 值明显增加, 且下一跳变为 R1 的地址 (192.168.13.2)。这证明 R3 已自动重新计算出经由 R1-R2-R4 的备用路径。

```
C:\Users\Administrator>ping 192.168.2.10

正在 Ping 192.168.2.10 具有 32 字节的数据:
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124

192.168.2.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 0ms, 平均 = 0ms

C:\Users\Administrator>tracert 192.168.2.10

通过最多 30 个跃点跟踪到 192.168.2.10 的路由

 1      1 ms      <1 毫秒      <1 毫秒  192.168.1.1
 2      1 ms      1 ms      <1 毫秒  192.168.13.2
 3     <1 毫秒    <1 毫秒    <1 毫秒  192.168.12.2
 4      1 ms      <1 毫秒    <1 毫秒  192.168.24.2
 5     <1 毫秒    <1 毫秒    <1 毫秒  192.168.2.10

跟踪完成。
```

图 35. 故障后: PC1 ping PC2 依然成功, 且 tracert 显示路径已改变

```
C:\Users\Administrator>ping 192.168.1.10

正在 Ping 192.168.1.10 具有 32 字节的数据:
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124

192.168.1.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 0ms, 平均 = 0ms

C:\Users\Administrator>tracert 192.168.1.10

通过最多 30 个跃点跟踪到 192.168.1.10 的路由

 1    <1 毫秒    <1 毫秒    <1 毫秒  192.168.2.1
 2    <1 毫秒    <1 毫秒    <1 毫秒  192.168.24.1
 3     1 ms      1 ms      <1 毫秒  192.168.12.1
 4     1 ms     <1 毫秒    <1 毫秒  192.168.13.1
 5    <1 毫秒    <1 毫秒    <1 毫秒  192.168.1.10

跟踪完成。
```

图 36. 故障后: PC2 ping PC1 也依然成功, 且 tracert 显示路径也已改变

最终分析: 拔掉冗余链路后, ping 测试依然能够成功。对比故障前后的 tracert 结果, 路径已自动、快速地切换为经由 R1 和 R2 的备用路径。这完美地证明了 OSPF 协议的快速收敛和自愈能力。

2 拓展实验二: 多区域 OSPF 与认证

2.1 实验要求

本实验要求在给定的该实验的多区域网络拓扑下, 完成以下任务:

- (a) 在所有路由器上配置多区域 OSPF 协议, 要求根据拓扑图, 通过相应命令将网络划分为 Area 0 (骨干区域)、Area 1 和 Area 2。同时, 需要增加路由器之间的认证, 并最终实现跨区域主机之间的相互 ping 通。
- (b) 设计并实施有效方法捕获路由器之间的路由信息交换报文, 重点比较并分析在多

区域环境下，不同区域之间（Inter-Area）与区域内部（Intra-Area）的路由信息报文交换内容（特别是 LSA 类型）的差异。

2.2 实验拓扑、接线与 IP 规划

本实验在逻辑上将网络划分为 Area 0(骨干区域)、Area 1 和 Area 2 三个区域，其中 R1 和 R2 作为 ABR（区域边界路由器）。

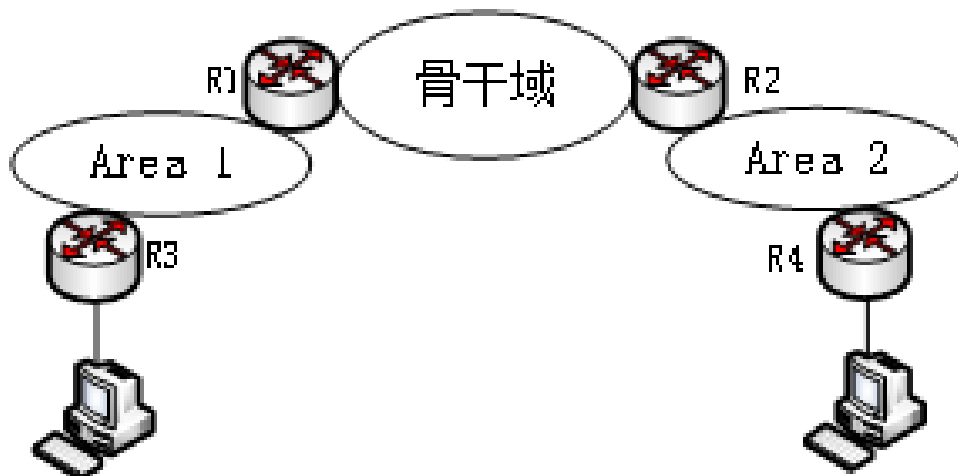


图 37. 拓展实验二网络拓扑图

- **物理接线:** (沿用拓展实验一的接线，但不使用 R3-R4 的冗余链路)

- PC1(实验网卡) ↔ R3 G0/0/1
- PC2(实验网卡) ↔ R4 G0/0/7
- R3 G0/0/0 ↔ R1 G0/0/0
- R1 G0/0/1 ↔ R2 G0/0/7
- R2 G0/0/5 ↔ R4 G0/0/5
- PC3(实验网卡，用于抓包) ↔ R1 G0/0/2 (观察端口)

- **IP 规划:** (沿用拓展实验一的 IP 规划)

- PC1-R3: 192.168.1.0/24
- PC2-R4: 192.168.2.0/24
- R3-R1: 192.168.13.0/24
- R1-R2: 192.168.12.0/24
- R2-R4: 192.168.24.0/24

2.3 实验步骤与结果分析

操作描述: 首先, 在所有路由器上执行 `undo ospf 1` 命令, 清除拓展实验一的配置。然后, 为捕获 OSPF 报文进行端口镜像配置, 并在 PC3 上重新启动 Wireshark 进行抓包。最后, 按照 R1、R2、R3、R4 的顺序依次配置多区域 OSPF, 以观察网络逐步建立的过程。

2.3.1 配置端口镜像

操作描述: 为了捕获并比较不同区域的 OSPF 报文, 我们在作为 ABR 的 R1 上配置端口镜像。R1 作为 Area 0 和 Area 1 的交界点, 是最佳的观测位置。我们将监听连接 R3(Area 1) 的 G0/0/0 接口和连接 R2(Area 0) 的 G0/0/1 接口, 并将所有流量复制到连接 PC3 的 G0/0/2 观察端口。(如果想观察其他的路由器之间的路由信息交换报文, 可以采用完全类似的方式来配置镜像并观察。)

```
system-view
! 1. 定义G0/0/2为观察端口
observe-port interface GigabitEthernet 0/0/2
!
! 2. 监听Area 1的对话 (R1 <-> R3)
interface GigabitEthernet 0/0/0
mirror to observe-port both
quit
!
! 3. 监听Area 0的对话 (R1 <-> R2)
interface GigabitEthernet 0/0/1
mirror to observe-port both
quit
!
save
```

Listing 10. R1 端口镜像配置命令

2.3.2 多区域 OSPF 配置与抓包分析

1. 配置 R1 并观察: 首先配置作为 ABR 的 R1, 启动 OSPF 进程, 并分别在 Area 1 和 Area 0 中宣告相应网段。


```
2025-12-17 08:35:52 /home/mobaxterm telnet 172.16.15.203
Connected to 172.16.15.203.
Entering character mode.
Escape character is '^'.

Login authentication

Username:user
Password:

-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.15.3
Time      : 2025-12-17 00:29:27+00:00
-----

<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]ospf 1
[R1-ospf-1]area 1
[R1-ospf-1-area-0.0.0.1]network 192.168.13.0 0.0.0.255
[R1-ospf-1-area-0.0.0.1]quit
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]ospf authentication-mode simple plain gda1pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R1-GigabitEthernet0/0/0]quit
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]ospf authentication-mode simple plain gda0pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R1-GigabitEthernet0/0/1]quit
[R1]quit
<R1>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 38. R1 (ABR) OSPF 与认证配置过程

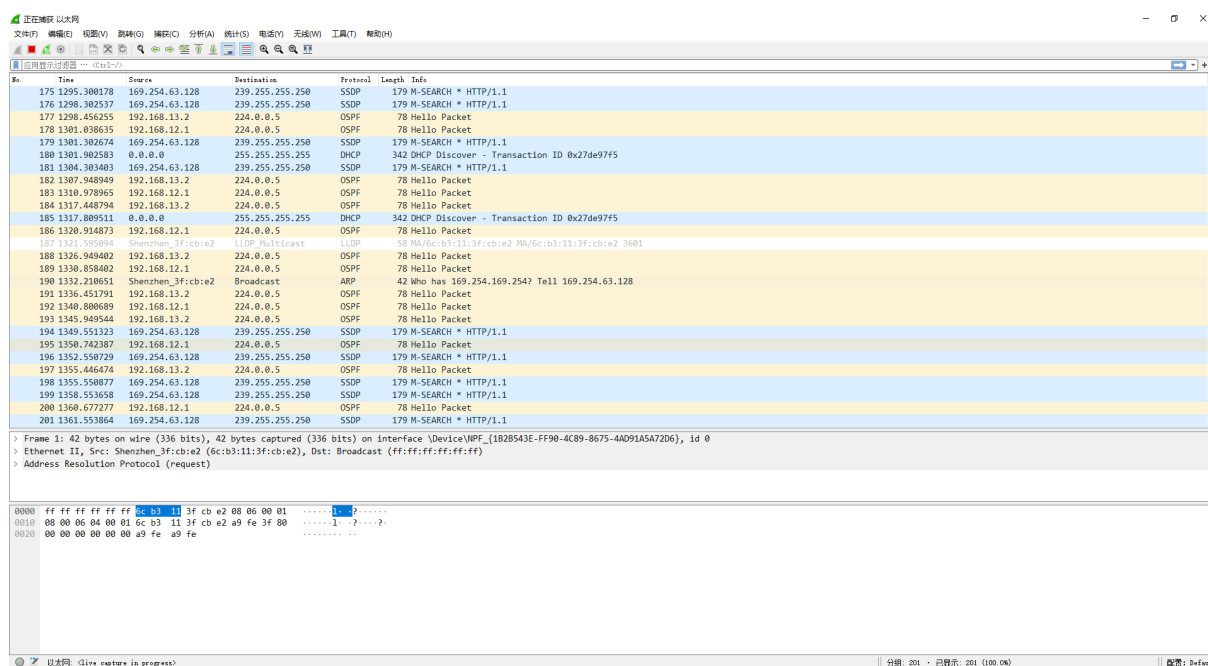


图 39. 配置 R1 后，Wireshark 捕获到 R1 向其所在两个区域发送的 Hello 报文

分析：配置完成后，R1 开始在其激活的接口上周期性地发送 Hello 报文，尝试发现邻居。此时网络中没有其他 OSPF 路由器，因此只能看到 R1 单方面“呼叫”的报文。

2. 配置 R2 并观察：接着配置另一台 ABR 路由器 R2。

```
2025-12-17 08:41:47 /home/mobaxterm telnet 172.16.15.204
Connected to 172.16.15.204.
Entering character mode
Escape character is '^'.

Login authentication

Username:user
Password:
-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.15.3
Time      : 2025-12-16 22:26:55+00:00
-----

<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]ospf 1 router-id 2.2.2.2
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]quit
[R2-ospf-1]area 2
[R2-ospf-1-area-0.0.0.2]network 192.168.24.0 0.0.0.255
[R2-ospf-1-area-0.0.0.2]quit
[R2-ospf-1]quit
[R2]interface GigabitEthernet0/0/7
[R2-GigabitEthernet0/0/7]ospf authentication-mode simple plain gda0pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R2-GigabitEthernet0/0/7]quit
[R2]interface GigabitEthernet0/0/5
[R2-GigabitEthernet0/0/5]ospf authentication-mode simple plain gda2pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R2-GigabitEthernet0/0/5]quit
[R2]quit
<R2>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 40. R2 (ABR) OSPF 与认证配置过程

No.	Time	Source	Destination	Protocol	Length	Info
286	1658.417089	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2adacbd0
287	1658.870447	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
288	1658.938666	HuaweiTe_70:d0:e6	Broadcast	ARP	60	Who has 192.168.12.1? Tell 192.168.12.2
289	1658.938907	HuaweiTe_70:d0:e6	Broadcast	ARP	60	Who has 192.168.12.1? Tell 192.168.12.2
290	1658.940004	64:bf:6b:a9:2c:83	HuaweiTe_70:d0:e6	ARP	60	192.168.12.1 is at 64:bf:6b:a9:2c:83
291	1658.940724	192.168.12.2	192.168.12.1	OSPF	66	OS Description
292	1658.942284	192.168.12.1	192.168.12.2	OSPF	66	OS Description
293	1658.942313	192.168.12.1	192.168.12.2	OSPF	106	OS Description
294	1658.943214	192.168.12.2	192.168.12.1	OSPF	82	LS Request
295	1658.943383	192.168.12.2	192.168.12.1	OSPF	106	OS Description
296	1658.945405	64:bf:6b:a9:2c:83	HuaweiTe_70:d0:e6	ARP	60	192.168.12.1 is at 64:bf:6b:a9:2c:83
297	1658.966841	192.168.12.1	192.168.12.2	OSPF	126	LS Update
298	1658.966841	192.168.12.1	192.168.12.2	OSPF	82	LS Request
299	1658.966841	192.168.12.1	192.168.12.2	OSPF	66	OS Description
300	1658.968023	192.168.12.2	192.168.12.1	OSPF	126	LS Update
301	1658.970012	192.168.12.2	224.0.0.5	OSPF	194	LS Update
302	1658.980395	192.168.12.1	224.0.0.5	OSPF	98	LS Update
303	1658.980953	192.168.12.2	224.0.0.5	OSPF	98	LS Update
304	1659.455547	192.168.13.2	224.0.0.5	OSPF	78	Hello Packet
305	1659.551619	192.168.12.1	192.168.12.2	OSPF	78	LS Acknowledge
306	1659.581619	192.168.12.1	224.0.0.5	OSPF	138	LS Acknowledge
307	1660.135115	fe80::391a:9152:ab9...ff02::1:2		DHCPv6	148	Solicit XID: 0xdff368 CID: 0001000130d3ae7010e7c63b2ade
308	1666.062389	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
309	1666.908102	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2adacbd0
310	1668.135206	fe80::391a:9152:ab9...ff02::1:2		DHCPv6	148	Solicit XID: 0xdff368 CID: 0001000130d3ae7010e7c63b2ade
311	1668.813462	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
312	1668.814469	192.168.12.2	224.0.0.5	OSPF	138	LS Update
313	1668.954127	192.168.13.2	224.0.0.5	OSPF	78	Hello Packet
314	1669.345571	192.168.12.1	224.0.0.5	OSPF	82	LS Acknowledge
315	1675.559351	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
316	1678.458843	192.168.13.2	224.0.0.5	OSPF	78	Hello Packet
317	1678.754095	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
318	1682.906573	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2adacbd0
319	1684.135716	fe80::391a:9152:ab9...ff02::1:2		DHCPv6	148	Solicit XID: 0xdff368 CID: 0001000130d3ae7010e7c63b2ade

> Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{1828543E-FF90-4C89-8675-AD091A5A7206}, Id 0
> Ethernet II, Src: Shenzhen_3f:cb:e2 (6c:b3:11:3f:cb:e2), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)

0000 ff ff ff ff ff ff ff ff 11 3f cb e2 08 06 00 011?.....
0010 08 06 04 00 01 6c b3 11 3f cb e2 a9 fe 3f 801?....?
0020 00 00 00 00 00 a9 fe a9 fe

图 41. 配置 R2 后, R1 与 R2 在骨干区域 Area 0 建立邻居关系的全过程

分析: R2 启动 OSPF 后, R1 与 R2 在共同的骨干区域 Area 0 内通过完整的五种报文交换 (Hello, DD, LSR, LSU, LSAck) 建立了 ‘Full’ 状态的邻居关系。下面是对其中关键报文的详细分析:

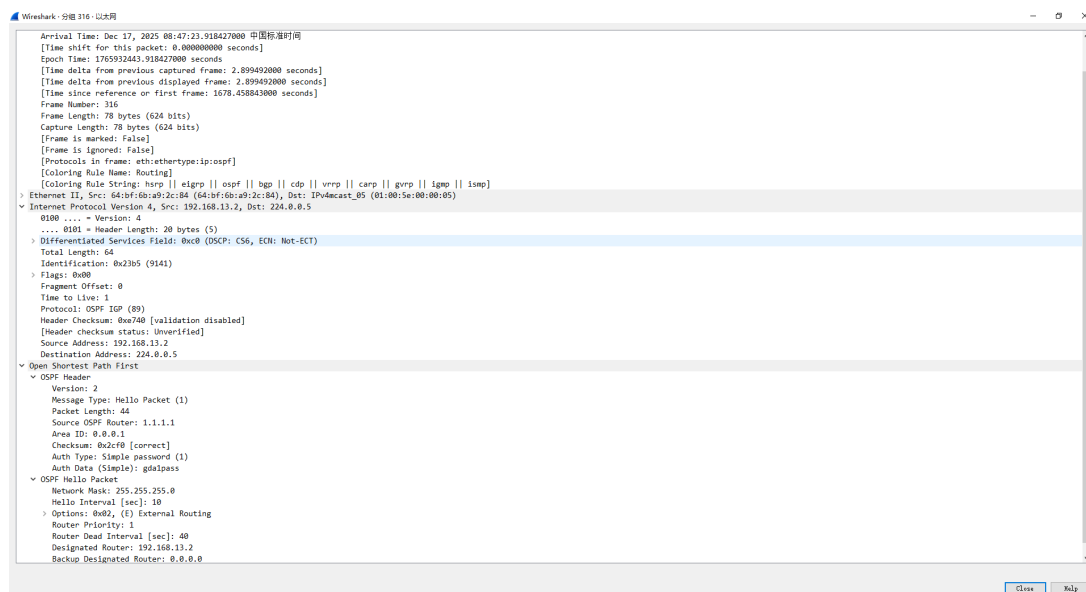


图 42. Hello 报文 (Type 1) 详情

Hello 报文分析: 如图所示, Hello 报文用于发现、建立和维护邻居关系。展开其详情, 可以看到 Area ID: 0.0.0.0 (Backbone), 表明这是一个骨干区域的问候; Auth Type: Simple password 和 Auth Data 显示了我们配置的认证信息, 只有密码匹配的路由器才能成为邻居; Router Dead Interval (40 秒) 定义了邻居失效的超时时间。两台路由器必须在这些核心参数上达成一致, 才能继续下一步。

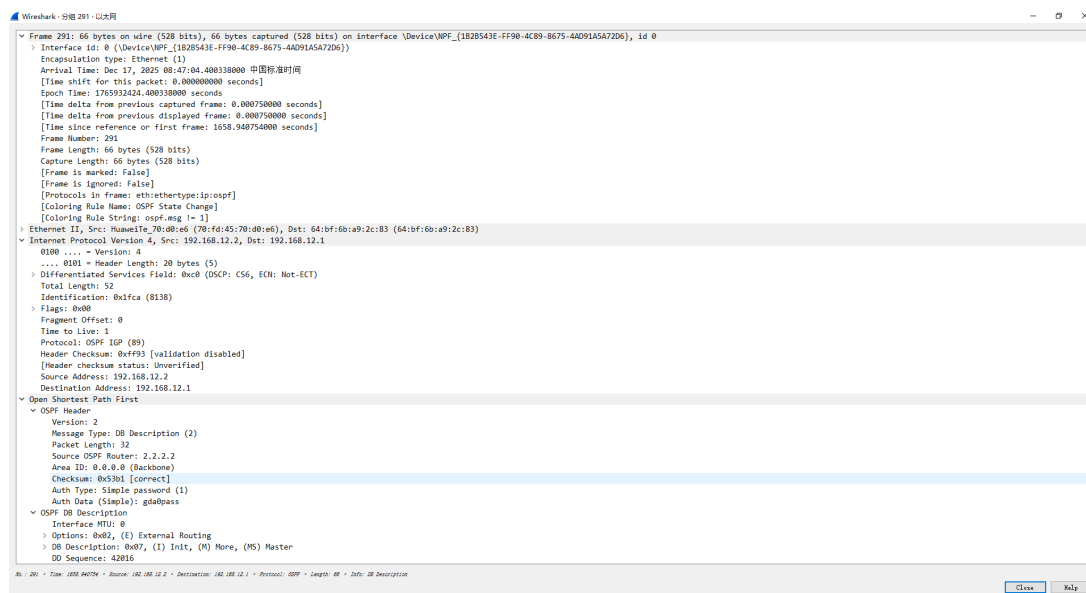


图 43. DB Description 报文 (Type 2) 详情

DB Description (DD) 报文分析: 在通过 Hello 包互相认识后, 路由器进入 ExStart 状态, 通过交换 DD 报文来选举主从 (Master/Slave) 并准备同步数据库。图中这

个 DD 报文由 R2 (Source OSPF Router: 2.2.2.2) 发出, 其 DB Description 字段中的 (MS) Master 标志位被置 1, 表明在此次协商中, 拥有更高 Router ID 的 R2 成为了 Master, 负责决定数据库同步的初始序列号 (DD Sequence)。

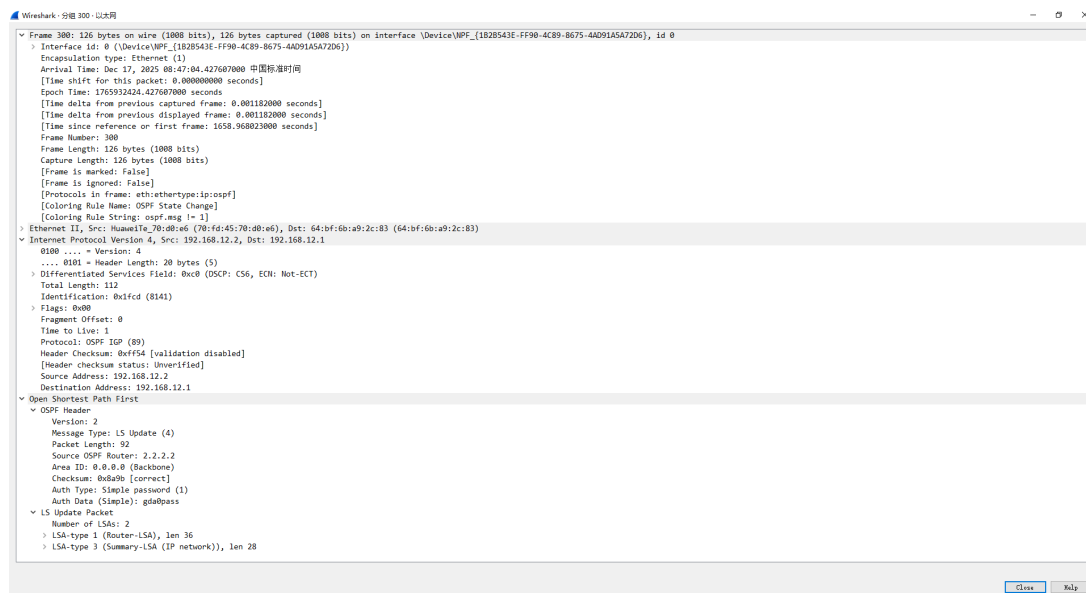


图 44. LS Update 报文 (Type 4) 详情

LS Update (LSU) 报文分析: LSU 报文是 OSPF 信息交换的核心, 它承载了网络的拓扑信息——LSA。如图所示, 这个 LSU 报文内部封装了两种 LSA: LSA-type 1 (Router-LSA) 和 LSA-type 3 (Summary-LSA)。

3. 配置 R3 并观察: 然后配置 Area 1 的内部路由器 R3。

```
2025-12-17 09:00:25 /home/mobaxterm telnet 172.16.17.203
connected to 172.16.17.203.
Entering character mode
Escape character is '^'.

Login authentication

Username:user
Password:

-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.17.3
Time : 2025-12-16 13:44:26+00:00
-----

<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]ospf 1 router-id 3.3.3.3
[R3-ospf-1]area 1
[R3-ospf-1-area-0.0.0.1]network 192.168.1.0 0.0.0.255
[R3-ospf-1-area-0.0.0.1]network 192.168.13.0 0.0.0.255
[R3-ospf-1-area-0.0.0.1]quit
[R3-ospf-1]quit
[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]ospf authentication-mode simple plain gda!pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R3-GigabitEthernet0/0/0]quit
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1]ospf authentication-mode simple plain gda!pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R3-GigabitEthernet0/0/1]quit
[R3]quit
<R3>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 45. R3 (Area 1 内部) OSPF 与认证配置过程

No.	Time	Source	Destination	Protocol	Length	Info
896	3072.352172	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
897	3075.086762	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
898	3075.072289	64:bf:6b:a8:b1:39	Broadcast	ARP	60	Who has 192.168.13.2? Tell 192.168.13.1
899	3075.072592	64:bf:6b:a8:b1:39	Broadcast	ARP	60	Who has 192.168.13.2? Tell 192.168.13.1
900	3075.073245	64:bf:6b:a9:2c:84	64:bf:6b:a8:b1:39	ARP	60	192.168.13.2 is at 64:bf:6b:a9:2c:84
901	3075.074528	192.168.13.2	192.168.13.1	OSPF	66	DB Description
902	3075.075806	192.168.13.2	192.168.13.1	OSPF	66	DB Description
903	3075.076479	192.168.13.2	192.168.13.1	OSPF	126	DB Description
904	3075.078880	64:bf:6b:a9:2c:84	64:bf:6b:a8:b1:39	ARP	60	192.168.13.2 is at 64:bf:6b:a9:2c:84
905	3075.080788	192.168.13.1	192.168.13.2	OSPF	94	LS Request
906	3075.083972	192.168.13.1	192.168.13.2	OSPF	86	DB Description
907	3075.088650	192.168.13.2	192.168.13.1	OSPF	154	LS Update
908	3075.098821	192.168.13.2	192.168.13.1	OSPF	70	LS Request
909	3075.098838	192.168.13.2	192.168.13.1	OSPF	66	DB Description
910	3075.110628	192.168.13.1	192.168.13.2	OSPF	118	LS Update
911	3075.119807	192.168.13.2	224.0.0.5	OSPF	98	LS Update
912	3075.153001	192.168.13.1	224.0.0.5	OSPF	234	LS Update
913	3075.496589	192.168.13.2	192.168.13.1	OSPF	98	LS Acknowledge
914	3075.496644	192.168.13.2	224.0.0.5	OSPF	118	LS Acknowledge
915	3076.218480	192.168.13.1	224.0.0.5	OSPF	98	LS Update
916	3076.811858	192.168.12.2	224.0.0.5	OSPF	78	LS Acknowledge
917	3080.295042	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
918	3081.857593	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
919	3082.627655	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
920	3084.554080	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
921	3084.585271	192.168.13.1	224.0.0.5	OSPF	142	LS Update
922	3085.497599	192.168.13.2	224.0.0.5	OSPF	98	LS Acknowledge
923	3090.233298	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
924	3091.354525	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
925	3091.972615	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
926	3094.069405	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
927	3100.179361	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
928	3100.853405	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
929	3101.915686	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet

Frame 291: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1828543E-FF90-4C89-8675-4A091A5A72D6}, id 0
Interface id: 0 (\Device\NPF_{1828543E-FF90-4C89-8675-4A091A5A72D6})
Encapsulation type: Ethernet (1)
Arrival Time: Dec 17, 2025 08:47:04.400338000 中国标准时间
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1765932424.400338000 seconds
[Time shift: 0.000000000 seconds]
0000 64 bf 6b a9 2c 83 70 fd 45 70 d0 05 08 00 45 c0 d k E:
0010 00 34 1f ca 00 00 01 59 ff 93 c0 a8 0c 02 c0 a8 4 Y
0020 0c 01 02 00 20 02 02 02 00 00 00 53 b1 S
0030 00 01 67 64 61 30 70 61 73 73 00 00 02 07 00 00 . g d a ! p a s s

图 46. 配置 R3 后，捕获到 R1 与 R3 在 Area 1 建立邻居关系的全过程

核心差异分析：比较区域内外 LSA 类型

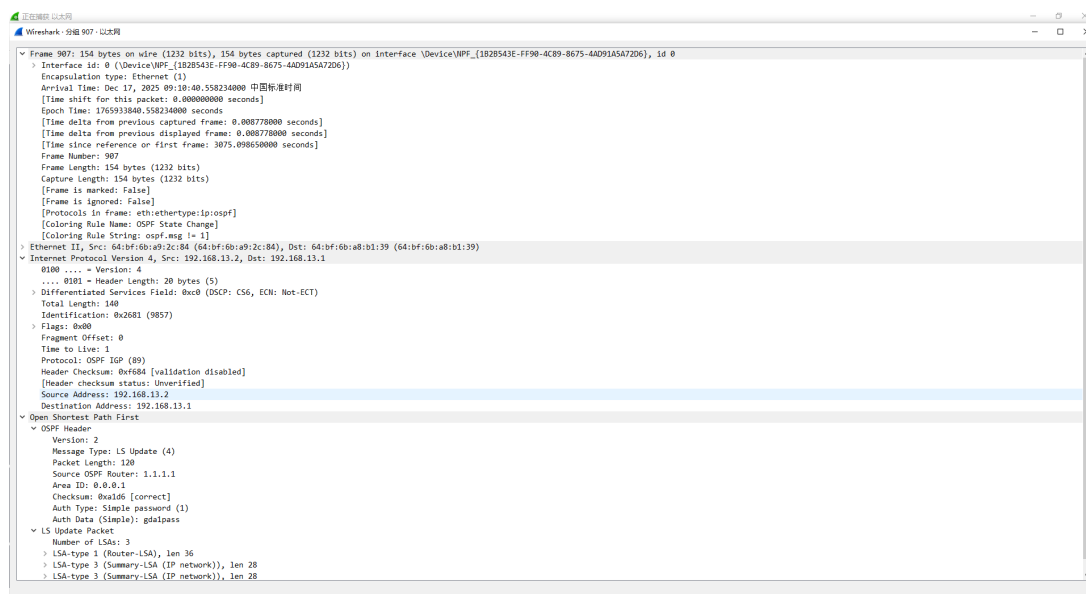


图 47. R1 与 R3 在 Area 1（区域内）交换的 LSU，封装了详细的 Type-1/2 LSA

分析: 在区域内部(如 R1-R3),路由器交换的是包含详细拓扑信息的 Type-1(Router-LSA) 和 Type-2(Network-LSA)。

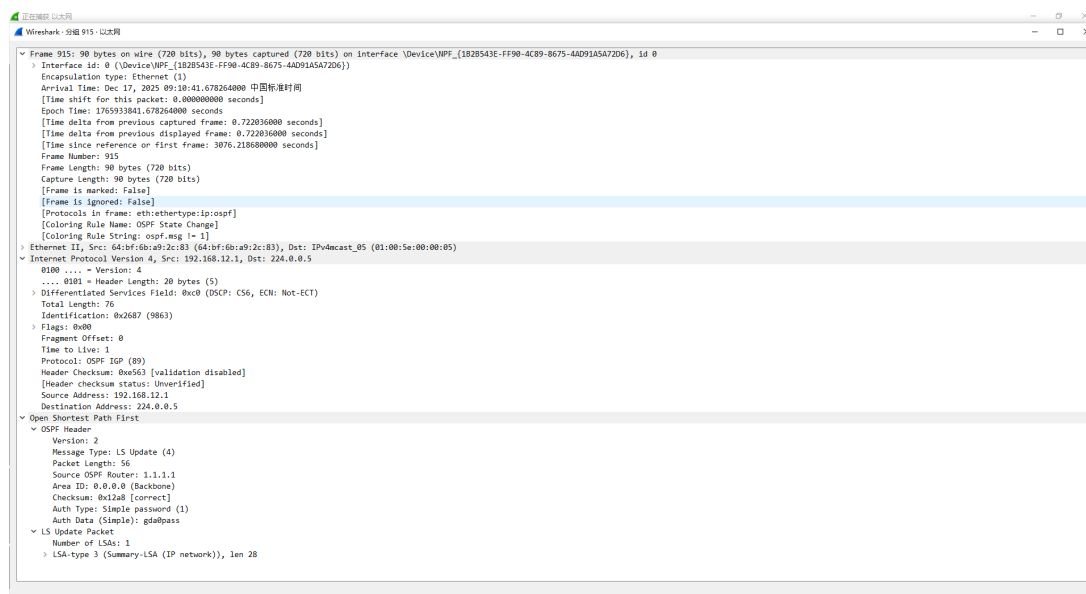


图 48. R1 向 R2 在 Area 0（跨区域）发送的 LSU，封装了精简的 Type-3 Summary LSA

分析: 当 ABR（R1）需要将 Area 1 的路由信息通告给骨干区（R2）时，它并不会直接转发原始的 Type-1/2 LSA，而是生成一条全新的、内容精简的 Type-3 Summary LSA 来通告路由摘要。通过对比这两张 LSU 截图，可以清晰地看出 OSPF 多区域设计的路由汇总和信息隐藏机制。

4. 配置 R4 并观察： 最后配置 Area 2 的内部路由器 R4。

```
2025-12-17 09:11:04 /home/mobaxterm telnet 172.16.17.204
Connected to 172.16.17.204.
Entering character mode
Escape character is '^'.

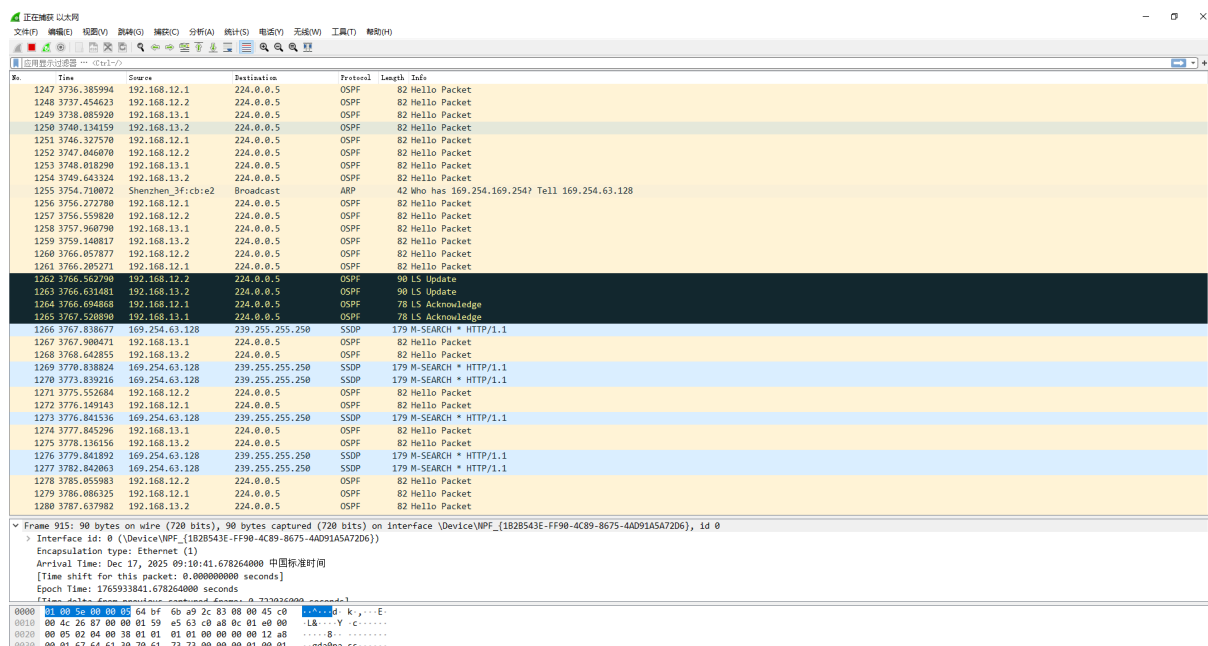
Login authentication

Username:user
Password:
-----
User last login information:
Access Type: Telnet
IP-Address : 172.16.17.3
Time      : 2025-12-16 22:12:27+00:00
-----

<R4>system-view
Enter system view, return user view with Ctrl+Z.
[R4]ospf 1 router-id 4.4.4.4
[R4-ospf-1]area 2
[R4-ospf-1-area-0.0.0.2]network 192.168.2.0 0.0.0.255
[R4-ospf-1-area-0.0.0.2]network 192.168.24.0 0.0.0.255
[R4-ospf-1-area-0.0.0.2]quit
[R4-ospf-1]quit
[R4]interface GigabitEthernet0/0/5
[R4-GigabitEthernet0/0/5]ospf authentication-mode simple plain gda2pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R4-GigabitEthernet0/0/5]quit
[R4]interface GigabitEthernet0/0/7
[R4-GigabitEthernet0/0/7]ospf authentication-mode simple plain gda2pass
Warning: The algorithm is not safe, and hmac-sha256 is recommended.
Warning: After plaintext mode is specified, the password will be saved in plaintext in the configuration file, which has a high security risk.
[R4-GigabitEthernet0/0/7]quit
[R4]quit
<R4>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 49. R4 (Area 2 内部) OSPF 与认证配置过程



No.	Time	Source	Destination	Protocol	Length	Info
1247	3736.385994	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1248	3737.454623	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1249	3738.085920	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
1250	3740.134158	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
1251	3746.327570	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1252	3747.046070	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1253	3748.018290	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
1254	3749.643324	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
1255	3754.710072	Shenzhen-3fc2be2	Broadcast	ARP	42	Who has 169.254.169.254? Tell 169.254.63.128
1256	3756.272780	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1257	3756.559820	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1258	3757.960790	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
1259	3759.140817	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
1260	3766.057877	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1261	3766.205271	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1262	3766.562790	192.168.12.2	224.0.0.5	OSPF	90	LS Update
1263	3766.631481	192.168.13.2	224.0.0.5	OSPF	90	LS Update
1264	3768.084080	192.168.12.1	224.0.0.5	OSPF	78	LS Acknowledge
1265	3767.520890	192.168.13.1	224.0.0.5	OSPF	78	LS Acknowledge
1266	3767.838677	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1267	3767.908471	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
1268	3768.642855	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
1269	3770.838824	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1270	3773.839216	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1271	3775.552684	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1272	3776.149143	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1273	3776.841536	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1274	3777.045296	192.168.13.1	224.0.0.5	OSPF	82	Hello Packet
1275	3778.136156	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet
1276	3779.841892	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1277	3782.842063	169.254.63.128	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1278	3785.055983	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
1279	3786.086325	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
1280	3787.637982	192.168.13.2	224.0.0.5	OSPF	82	Hello Packet

Frame 915: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on Interface \Device\NPF_{1B2B543E-FF90-4C89-8675-4AD91A5A7206}, id 0
Interface id: 0 (\Device\NPF_{1B2B543E-FF90-4C89-8675-4AD91A5A7206})
Encapsulation type: Ethernet (1)
Arrival Time: Dec 17, 2025 09:10:41.678264000 中国标准时间
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1765933841.678264000 seconds
[Time delta from previous captured frame: 0.733026000 seconds]
0000 00 00 00 00 00 00 64 bf 6b a9 2c 83 08 00 45 c0 -->Id, K, ...E:
0010 00 4c 25 17 00 01 59 e5 63 c0 a8 0c 01 40 00 --LS...Y...C...:
0020 00 05 02 04 00 38 01 01 01 01 00 00 00 00 12 a8 --...B...:.....:
0030 00 01 67 64 61 30 70 61 73 73 00 00 00 01 00 01 --gda0pa ss.....:

图 50. 配置 R4 后，骨干区域捕获到由 R2 新生成的 LSU 和 LSAck 报文

分析: 当网络已稳定, 最后加入 R4 时, R2 作为 ABR, 会将从 R4 学到的 Area 2 内部路由, 通过新的 Type-3 LSA 汇总并发布到骨干区域。我们在 R1 的镜像点可以捕获到 R2 发来的这些更新 (LS Update) 及 R1 的确认 (LS Acknowledge), 这体现了 OSPF 的增量更新机制。

2.3.3 最终状态验证

操作描述: 检查所有路由器最终的 OSPF 邻居和路由表状态, 并进行端到端连通性测试。

```
[R1]display ospf peer

      OSPF Process 1 with Router ID 1.1.1.1
        Neighbors

Area 0.0.0.0 interface 192.168.12.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 2.2.2.2          Address: 192.168.12.2
  State: Full Mode:Nbr is Master Priority: 1
  DR: 192.168.12.2 BDR: 192.168.12.1 MTU: 0
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00:43:41
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.1 interface 192.168.13.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 3.3.3.3          Address: 192.168.13.1
  State: Full Mode:Nbr is Master Priority: 1
  DR: 192.168.13.1 BDR: 192.168.13.2 MTU: 0
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00:20:05
  Authentication Sequence: [ 0 ]

[R1]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
  127.0.0.0/8       Direct  0    0       D    127.0.0.1       InLoopBack0
  127.0.0.1/32      Direct  0    0       D    127.0.0.1       InLoopBack0
 127.255.255.255/32 Direct  0    0       D    127.0.0.1       InLoopBack0
  172.16.0.0/16     Direct  0    0       D    172.16.15.203   GigabitEthernet0/0/8
  172.16.15.203/32  Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/8
 172.16.255.255/32 Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/8
  192.168.1.0/24    OSPF    10    2       D    192.168.13.1    GigabitEthernet0/0/0
  192.168.2.0/24    OSPF    10    3       D    192.168.12.2    GigabitEthernet0/0/1
  192.168.12.0/24   Direct  0    0       D    192.168.12.1    GigabitEthernet0/0/1
  192.168.12.1/32   Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/1
 192.168.12.255/32 Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/1
  192.168.13.0/24   Direct  0    0       D    192.168.13.2    GigabitEthernet0/0/0
  192.168.13.2/32   Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/0
 192.168.13.255/32 Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/0
  192.168.24.0/24   OSPF    10    2       D    192.168.12.2    GigabitEthernet0/0/1
255.255.255.255/32 Direct  0    0       D    127.0.0.1       InLoopBack0
```

图 51. R1 最终邻居与路由表


```
[R2]display ospf peer

      OSPF Process 1 with Router ID 2.2.2.2
        Neighbors

Area 0.0.0.0 interface 192.168.12.2(GigabitEthernet0/0/7)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.12.1
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 192.168.12.2  BDR: 192.168.12.1  MTU: 0
  Dead timer due in 31 sec
  Retrans timer interval: 5
  Neighbor is up for 00:44:20
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.2 interface 192.168.24.1(GigabitEthernet0/0/5)'s neighbors
Router ID: 4.4.4.4      Address: 192.168.24.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.24.2  BDR: 192.168.24.1  MTU: 0
  Dead timer due in 38 sec
  Retrans timer interval: 0
  Neighbor is up for 00:09:13
  Authentication Sequence: [ 0 ]

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
127.0.0.0/8         Direct   0    0        D    127.0.0.1          InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1          InLoopBack0
127.255.255.255/32  Direct   0    0        D    127.0.0.1          InLoopBack0
172.16.0.0/16       Direct   0    0        D    172.16.15.204      GigabitEthernet0/0/9
172.16.15.204/32    Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/9
192.168.1.0/24       OSPF     10    3        D    192.168.12.1       GigabitEthernet0/0/7
192.168.2.0/24       OSPF     10    2        D    192.168.24.2       GigabitEthernet0/0/5
192.168.12.0/24     Direct   0    0        D    192.168.12.2       GigabitEthernet0/0/7
192.168.12.2/32     Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/7
192.168.12.255/32   Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/7
192.168.13.0/24     OSPF     10    2        D    192.168.12.1       GigabitEthernet0/0/7
192.168.24.0/24     Direct   0    0        D    192.168.24.1       GigabitEthernet0/0/5
192.168.24.1/32     Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/5
192.168.24.255/32   Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/5
255.255.255.255/32  Direct   0    0        D    127.0.0.1          InLoopBack0
```

图 52. R2 最终邻居与路由表

```

[R3]display ospf peer

      OSPF Process 1 with Router ID 3.3.3.3
        Neighbors

Area 0.0.0.1 interface 192.168.13.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 192.168.13.2
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.13.1 BDR: 192.168.13.2 MTU: 0
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00:17:29
Authentication Sequence: [ 0 ]

[R3]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
127.0.0.0/8         Direct  0    0       D    127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0    0       D    127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0       D    127.0.0.1       InLoopBack0
172.16.0.0/16        Direct  0    0       D    172.16.17.203   GigabitEthernet0/0/8
172.16.17.203/32     Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/8
172.16.255.255/32    Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/8
192.168.1.0/24        Direct  0    0       D    192.168.1.1     GigabitEthernet0/0/1
192.168.1.1/32        Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/1
192.168.1.255/32     Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/1
192.168.2.0/24        OSPF    10    4       D    192.168.13.2    GigabitEthernet0/0/0
192.168.12.0/24       OSPF    10    2       D    192.168.13.2    GigabitEthernet0/0/0
192.168.13.0/24       Direct  0    0       D    192.168.13.1    GigabitEthernet0/0/0
192.168.13.1/32       Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/0
192.168.13.255/32     Direct  0    0       D    127.0.0.1       GigabitEthernet0/0/0
192.168.24.0/24       OSPF    10    3       D    192.168.13.2    GigabitEthernet0/0/0
255.255.255.255/32    Direct  0    0       D    127.0.0.1       InLoopBack0

```

图 53. R3 最终邻居与路由表

```
[R4]display ospf peer

      OSPF Process 1 with Router ID 4.4.4.4
        Neighbors

Area 0.0.0.2 interface 192.168.24.2(GigabitEthernet0/0/5)'s neighbors
Router ID: 2.2.2.2      Address: 192.168.24.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 192.168.24.2 BDR: 192.168.24.1 MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:06:44
Authentication Sequence: [ 0 ]

[R4]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 16          Routes : 16

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
127.0.0.0/8         Direct   0    0        D    127.0.0.1          InLoopBack0
127.0.0.1/32        Direct   0    0        D    127.0.0.1          InLoopBack0
127.255.255.255/32  Direct   0    0        D    127.0.0.1          InLoopBack0
172.16.0.0/16       Direct   0    0        D    172.16.17.204      GigabitEthernet0/0/9
172.16.17.204/32    Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/9
172.16.255.255/32   Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/9
192.168.1.0/24       OSPF     10    4        D    192.168.24.1       GigabitEthernet0/0/5
192.168.2.0/24       Direct   0    0        D    192.168.2.1       GigabitEthernet0/0/7
192.168.2.1/32       Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/7
192.168.2.255/32     Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/7
192.168.12.0/24      OSPF     10    2        D    192.168.24.1       GigabitEthernet0/0/5
192.168.13.0/24      OSPF     10    3        D    192.168.24.1       GigabitEthernet0/0/5
192.168.24.0/24      Direct   0    0        D    192.168.24.2       GigabitEthernet0/0/5
192.168.24.2/32      Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/5
192.168.24.255/32    Direct   0    0        D    127.0.0.1          GigabitEthernet0/0/5
255.255.255.255/32   Direct   0    0        D    127.0.0.1          InLoopBack0
```

图 54. R4 最终邻居与路由表

状态分析: 从各路由器最终的 `display ospf peer` 和 `display ip routing-table` 输出中可以看到，网络已成功收敛。`display ospf peer` 的输出显示，每台路由器都与其物理直连的邻居成功建立起了‘Full’状态的邻接关系。`display ip routing-table` 的输出则显示，每台路由器都通过 OSPF 学习到了全网所有非直连网段的路由信息。例如，在 R3 的路由表中，可以看到协议类型为 OSPF、开销值 (Cost) 为 4 的去往 PC2 网段 (192.168.2.0/24) 的路由。这证明了即使在多区域环境下，路由信息也已通过骨干区域正确传递，实现了全网路由的自动学习。

```
C:\Users\Administrator>ping 192.168.2.10

正在 Ping 192.168.2.10 具有 32 字节的数据:
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.2.10 的回复: 字节=32 时间<1ms TTL=124

192.168.2.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 0ms, 平均 = 0ms

C:\Users\Administrator>tracert 192.168.2.10

通过最多 30 个跃点跟踪到 192.168.2.10 的路由

 1    <1 毫秒      1 ms      1 ms    192.168.1.1
 2    <1 毫秒      1 ms      <1 毫秒  192.168.13.2
 3    <1 毫秒      <1 毫秒    <1 毫秒  192.168.12.2
 4    <1 毫秒      <1 毫秒    <1 毫秒  192.168.24.2
 5    <1 毫秒      <1 毫秒    <1 毫秒  192.168.2.10

跟踪完成。
```

图 55. 最终连通性测试: PC1 ping 并 tracert PC2

```
C:\Users\Administrator>ping 192.168.1.10

正在 Ping 192.168.1.10 具有 32 字节的数据:
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间=1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124
来自 192.168.1.10 的回复: 字节=32 时间<1ms TTL=124

192.168.1.10 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 0ms, 最长 = 1ms, 平均 = 0ms

C:\Users\Administrator>tracert 192.168.1.10

通过最多 30 个跃点跟踪到 192.168.1.10 的路由

 1    <1 毫秒    <1 毫秒    <1 毫秒  192.168.2.1
 2    <1 毫秒    <1 毫秒    <1 毫秒  192.168.24.1
 3     1 ms      <1 毫秒    <1 毫秒  192.168.12.1
 4     1 ms      <1 毫秒     1 ms    192.168.13.1
 5    <1 毫秒    <1 毫秒    <1 毫秒  192.168.1.10

跟踪完成。
```

图 56. 最终连通性测试: PC2 ping 并 tracert PC1

分析: 最终的双向 ping 和 tracert 测试均成功, 证明在多区域 OSPF 环境下, 全网已实现可靠的端到端连通。以 PC2 tracert PC1 为例, 其路径 192.168.2.1 (R4) → 192.168.24.1 (R2) → 192.168.12.1 (R1) → 192.168.13.1 (R3) → 192.168.1.10 (PC1) 清晰地展示了数据包如何跨越 Area 2、Area 0 和 Area 1 最终到达目的地。这个结果不仅验证了网络的连通性, 更直观地呈现了 OSPF 协议在多区域环境中计算出的最优路径。

3 拓展实验三：默认路由与互联网出口控制

3.1 实验要求

- (a) 本次实验要求在拓展实验二已实现全网互通的基础上，参考给定的目标拓扑，设计并实施一套解决方案。使内部网络的所有 PC，能够通过且仅通过路由器 R1，访问到外部的校园网及互联网。为保证测试环境的纯净，要求关闭 PC 上的校园网网卡。

3.2 实验拓扑

本次实验的实验拓扑如下：

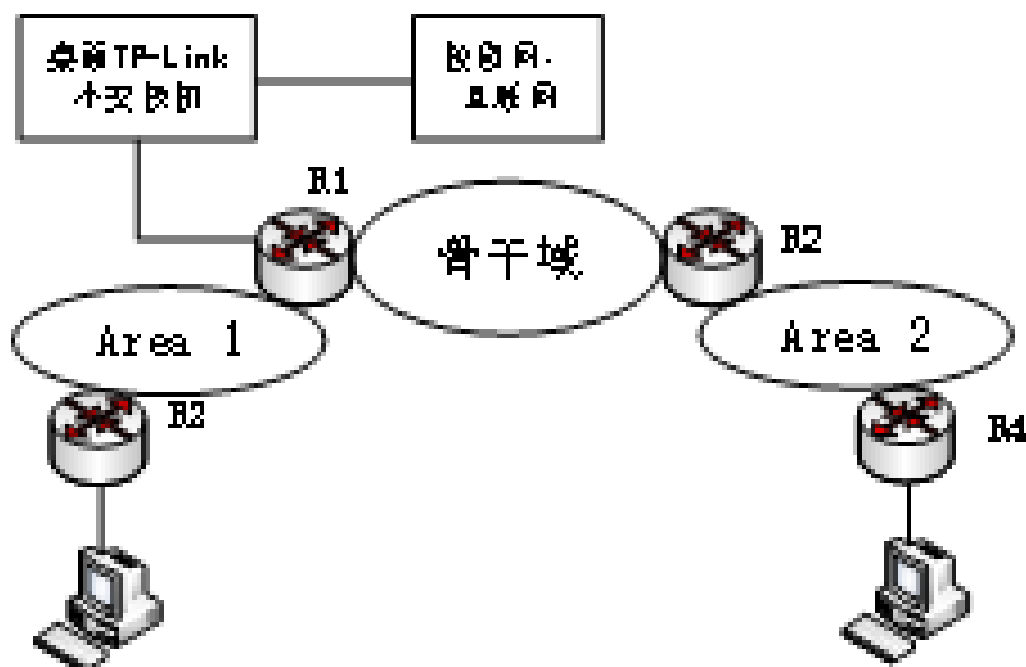


图 57. 拓展实验 3 网络拓扑图

3.3 实验步骤与结果分析

3.3.1 第一步：在 R1 上配置 NAT 与默认路由

操作描述：我们在作为出口的 R1 上配置 NAT（Easy IP）和一条指向校园网网关的静态默认路由，并通过 ‘default-route-advertise’ 命令将该默认路由注入到整个 OSPF 网络。

```
2025-12-17 09:36.35 /home/mobaxterm telnet 172.16.15.203
Connected to 172.16.15.203.
Entering character mode
Escape character is '^]'.

Login authentication

Username:user
Password:
-----
User last login information:
-----
Access Type: Telnet
IP-Address : 172.16.15.3
Time      : 2025-12-17 01:31:45+00:00
-----
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]acl number 2000
[R1-acl-basic-2000]rule 5 permit source 192.168.0.0 0.0.255.255
[R1-acl-basic-2000]quit
[R1]interface GigabitEthernet0/0/8
[R1-GigabitEthernet0/0/8]nat outbound 2000
[R1-GigabitEthernet0/0/8]quit
[R1]ip route-static 0.0.0.0 0 172.16.0.1
[R1]ospf 1
[R1-ospf-1]default-route-advertise always
[R1-ospf-1]quit
[R1]quit
<R1>quit

Info:Configuration console exit, please retry to log on
telnet: Connection closed by foreign host
```

图 58. 在 R1 上配置 NAT 和默认路由注入

3.3.2 第二步：验证默认路由的下发

操作描述：登录到区域内的路由器 R3 和 R4，查看它们的路由表，以验证默认路由是否已通过 OSPF 成功学习。


```
[R3]display ip routing-table
Route Flags: R - relay, D - download to fib, T - to vpn-instance
-----
Routing Tables: Public
Destinations : 17      Routes : 17
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	O_ASE	150	1	D	192.168.13.2	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/16	Direct	0	0	D	172.16.17.203	GigabitEthernet0/0/8
172.16.17.203/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/8
172.16.255.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/8
192.168.1.0/24	Direct	0	0	D	192.168.1.1	GigabitEthernet0/0/1
192.168.1.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.1.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.2.0/24	OSPF	10	4	D	192.168.13.2	GigabitEthernet0/0/0
192.168.12.0/24	OSPF	10	2	D	192.168.13.2	GigabitEthernet0/0/0
192.168.13.0/24	Direct	0	0	D	192.168.13.1	GigabitEthernet0/0/0
192.168.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.24.0/24	OSPF	10	3	D	192.168.13.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

图 59. R3 的路由表中出现了 O_ASE 类型的默认路由

```
[R4]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 17      Routes : 17
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	O_ASE	150	1	D	192.168.24.1	GigabitEthernet0/0/5
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/16	Direct	0	0	D	172.16.17.204	GigabitEthernet0/0/9
172.16.17.204/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/9
172.16.255.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/9
192.168.1.0/24	OSPF	10	4	D	192.168.24.1	GigabitEthernet0/0/5
192.168.2.0/24	Direct	0	0	D	192.168.2.1	GigabitEthernet0/0/7
192.168.2.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/7
192.168.2.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/7
192.168.12.0/24	OSPF	10	2	D	192.168.24.1	GigabitEthernet0/0/5
192.168.13.0/24	OSPF	10	3	D	192.168.24.1	GigabitEthernet0/0/5
192.168.24.0/24	Direct	0	0	D	192.168.24.2	GigabitEthernet0/0/5
192.168.24.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/5
192.168.24.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/5
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

图 60. R4 的路由表中也出现了 O_ASE 类型的默认路由

分析: 从 R3 和 R4 的路由表中, 可以清晰地看到多出了一条目标为 0.0.0.0/0 的路由。其协议类型为 O_ASE (OSPF 外部路由), 优先级为 150, 下一跳均指向去往 R1 的方向。这证明 R1 已成功将默认路由发布至全网。

3.3.3 第三步：最终互联网访问验证

操作描述：在 PC2 上进行真实的互联网访问测试，包括 ping、tracert 和浏览器访问。

```
C:\Users\Administrator>ping 202.116.64.8

正在 Ping 202.116.64.8 具有 32 字节的数据:
来自 202.116.64.8 的回复: 字节=32 时间=4ms TTL=245
来自 202.116.64.8 的回复: 字节=32 时间=2ms TTL=245
来自 202.116.64.8 的回复: 字节=32 时间=2ms TTL=245
来自 202.116.64.8 的回复: 字节=32 时间=3ms TTL=245

202.116.64.8 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 2ms, 最长 = 4ms, 平均 = 2ms

C:\Users\Administrator>tracert 202.116.64.8

通过最多 30 个跃点跟踪到 202.116.64.8 的路由

  1    <1 毫秒    <1 毫秒    <1 毫秒  192.168.2.1
  2    <1 毫秒    <1 毫秒    <1 毫秒  192.168.24.1
  3     1 ms     <1 毫秒    <1 毫秒  192.168.12.1
  4    <1 毫秒     1 ms     1 ms    172.16.0.1
  5      *        *        *        请求超时。
  6     3 ms     3 ms     3 ms    172.18.179.254
  7     4 ms     4 ms     4 ms    10.44.185.201
  8     4 ms     2 ms     1 ms    10.44.32.201
  9     3 ms     2 ms     1 ms    10.44.16.201
 10     6 ms     3 ms     2 ms    10.66.16.206
 11     4 ms     1 ms     2 ms    10.66.50.203
 12     2 ms     3 ms     2 ms    202.116.64.8

跟踪完成。
```

图 61. PC2 ping 并 tracert 中山大学官网 IP 地址

分析：ping 公网 IP 成功，证明了从 PC2 到互联网的数据通路已经打通，NAT 转换功能正常。tracert 的结果显示，数据流在内部网络中经由 R4、R2，最终汇聚到 R1 (192.168.12.1)，并从 R1 流向公网，路径符合预期。

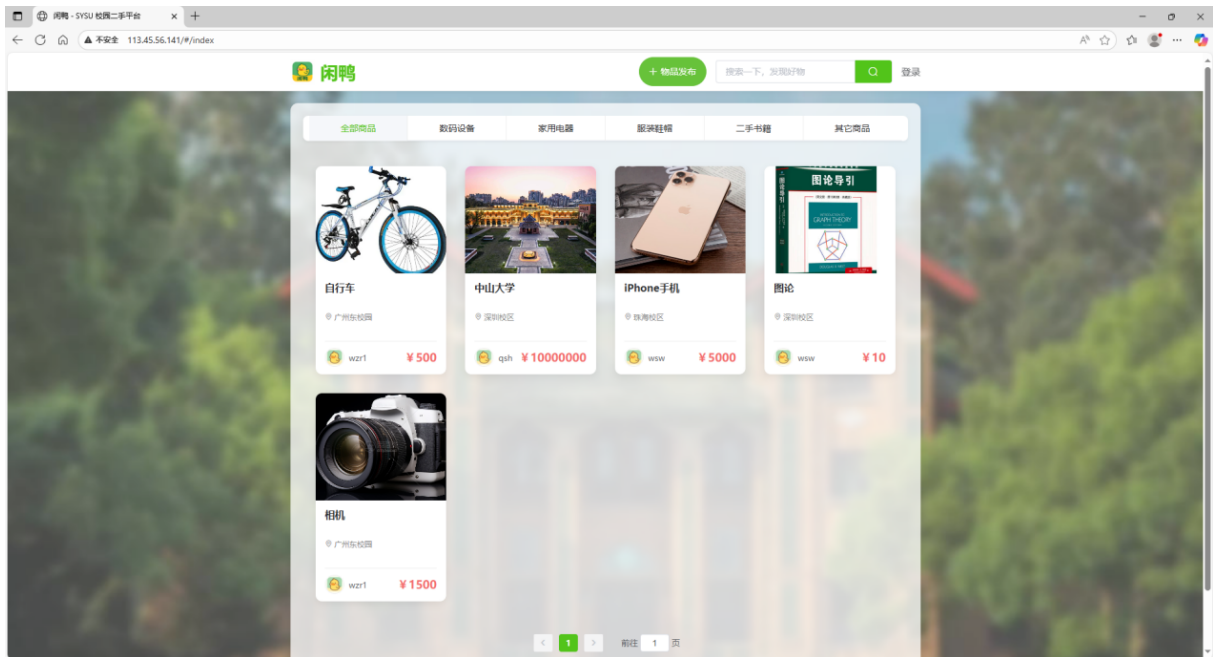


图 62. 通过 IP 地址成功访问笔者的个人网站

分析: 通过 IP 地址直接访问网站成功，进一步验证了 TCP 流量（HTTP）可以通过 NAT 设备正常访问公网。



图 63. 为 PC2 配置 DNS 服务器以实现域名解析

分析: 最初通过域名访问失败，是因为 PC 未配置 DNS 服务器。在手动配置了 DNS 服务器后，域名解析恢复正常。

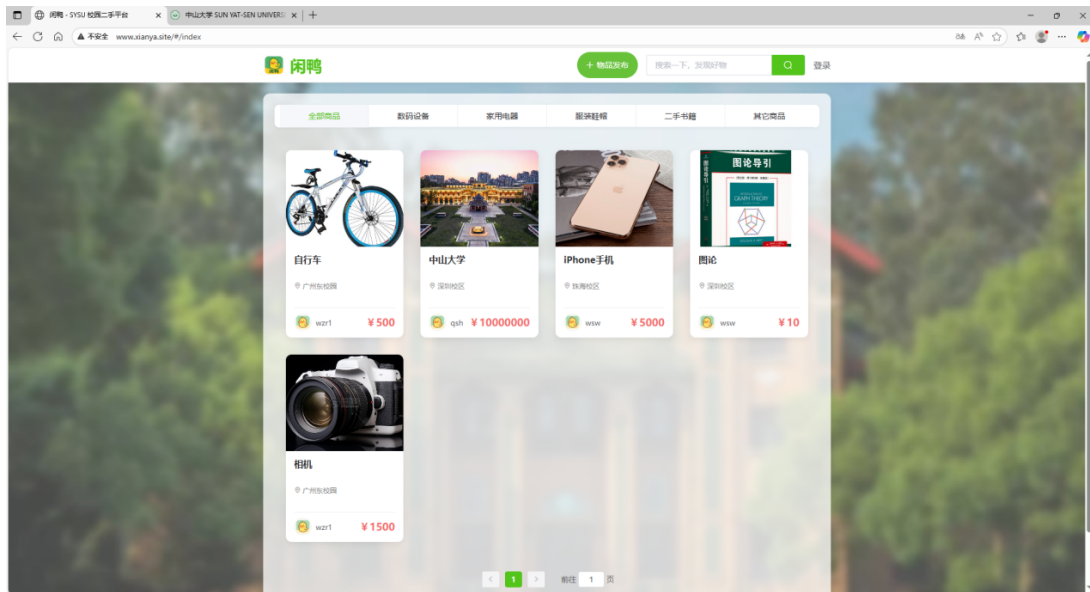


图 64. 通过域名成功访问笔者的个人网站

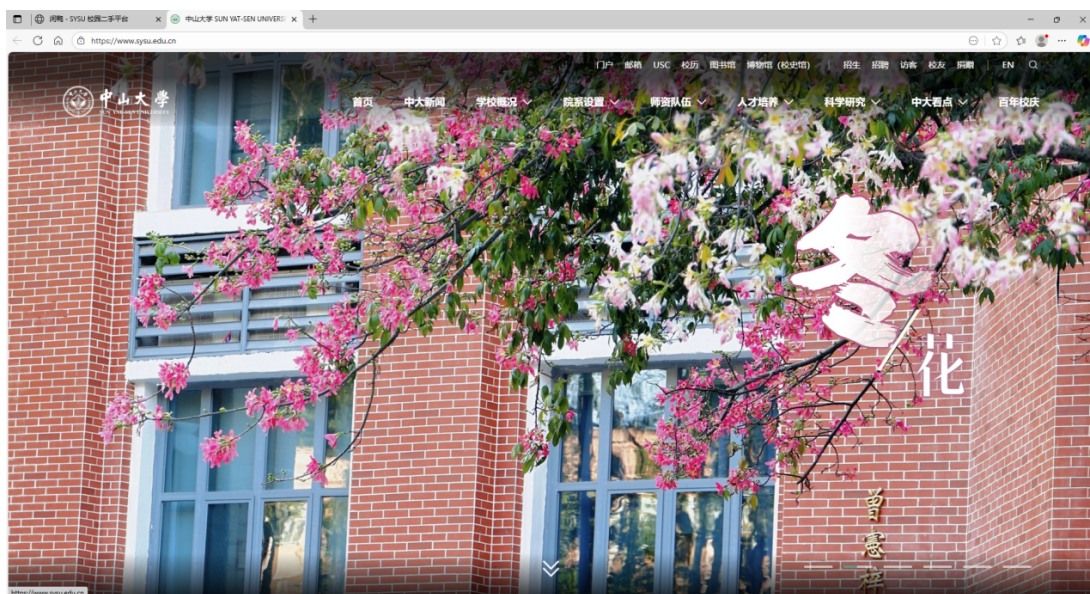


图 65. 通过域名成功访问中山大学官网

最终分析: 在解决了 DNS 问题后, PC2 已能通过域名自由访问互联网。至此, 我们成功地在多区域 OSPF 网络的基础上, 构建了一个由 R1 作为唯一 NAT 出口的、可访问真实互联网的网络环境, 拓展实验 3 圆满完成。